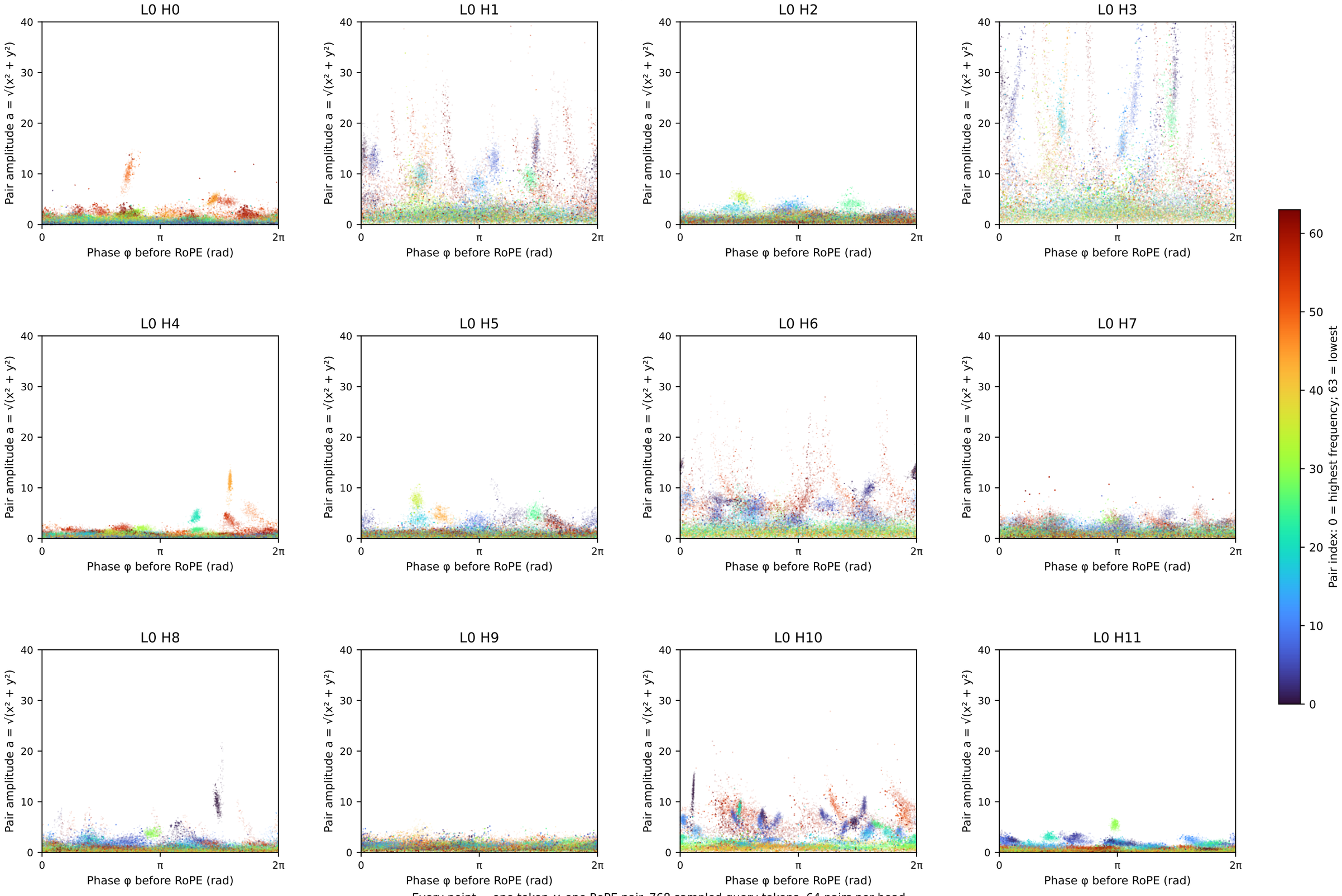
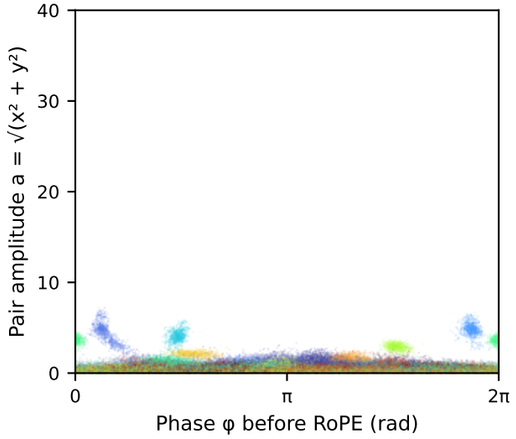
Qwen queries · layer 0 · amplitude axis 0–40



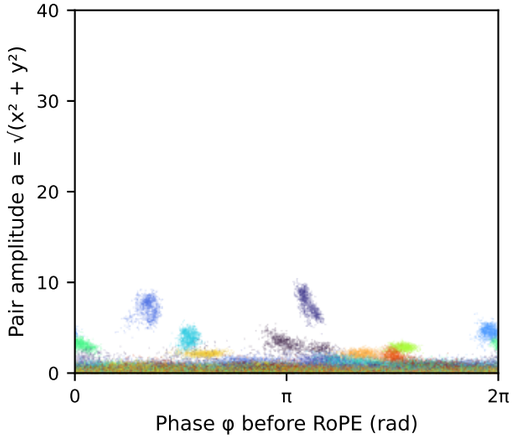
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 1 · amplitude axis 0–40

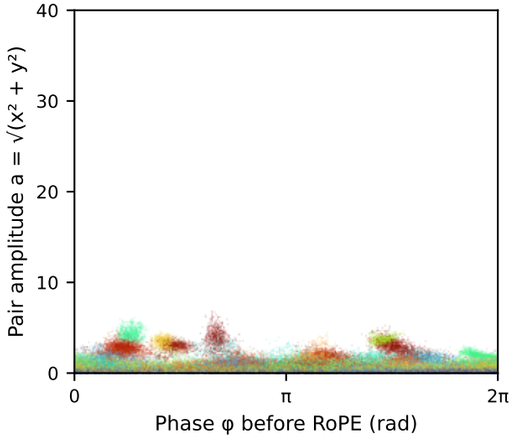
L1 H0



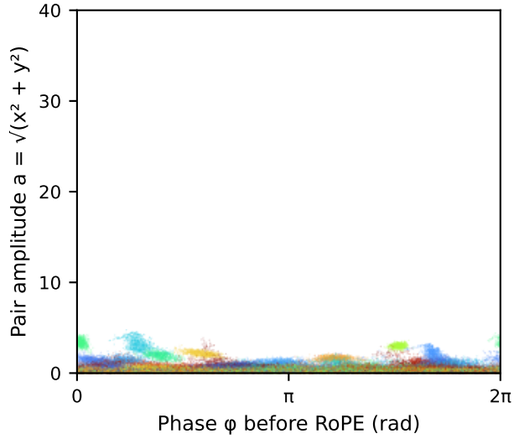
L1 H1



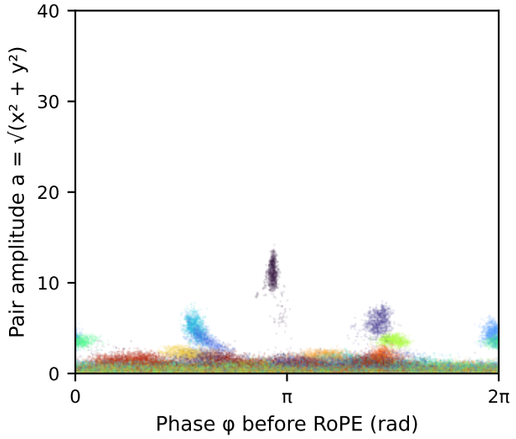
L1 H2



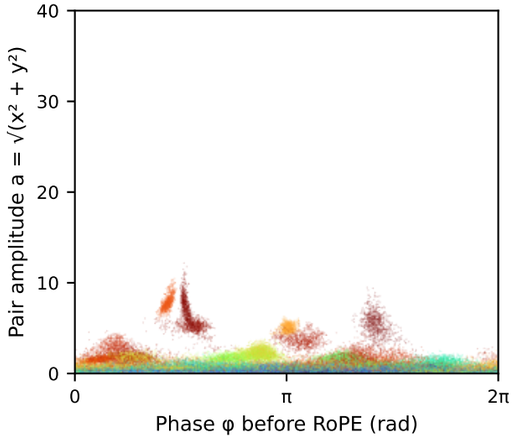
L1 H3



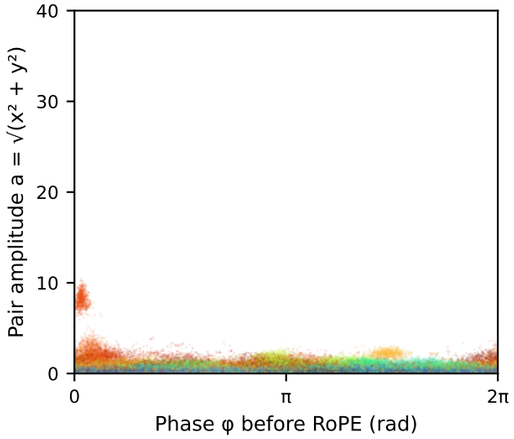
L1 H4



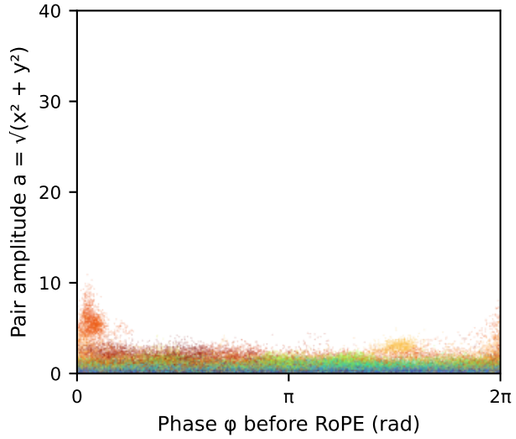
L1 H5



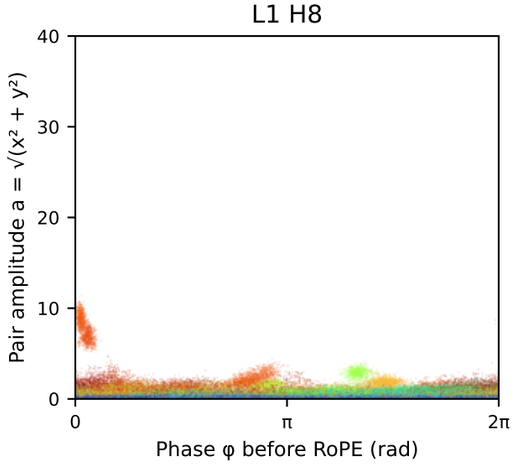
L1 H6



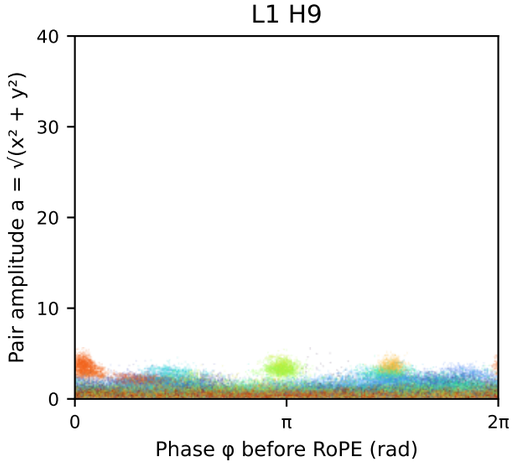
L1 H7



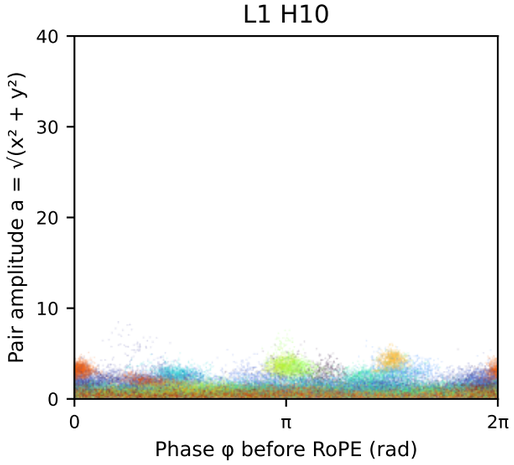
L1 H8



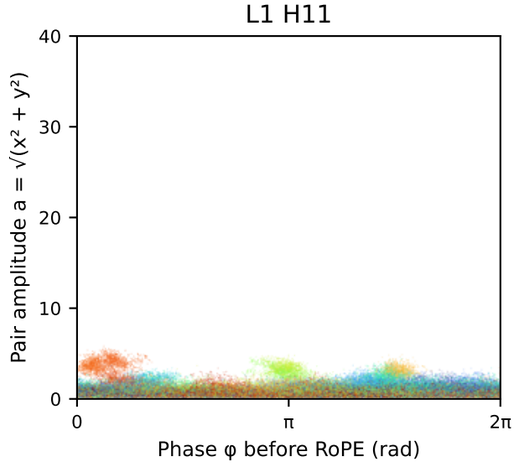
L1 H9

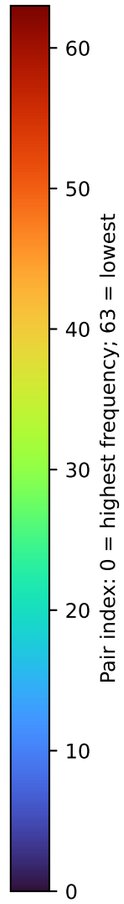


L1 H10



L1 H11

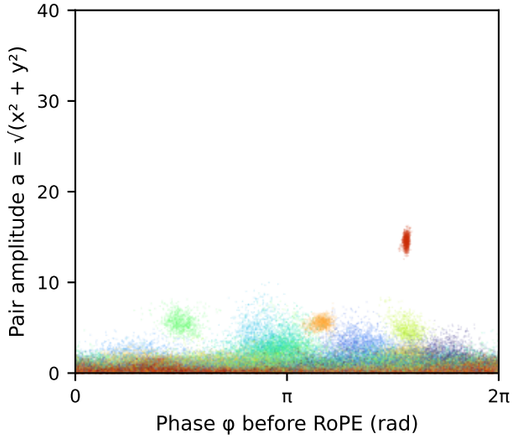




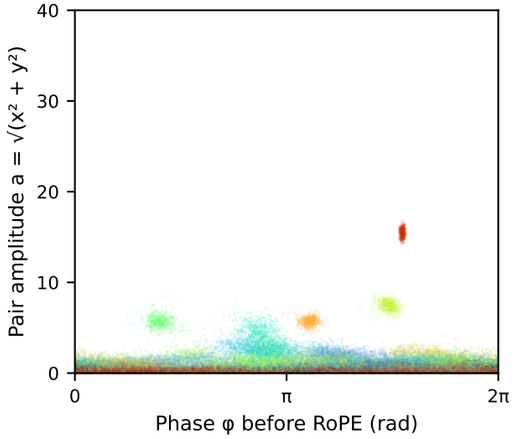
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 2 · amplitude axis 0–40

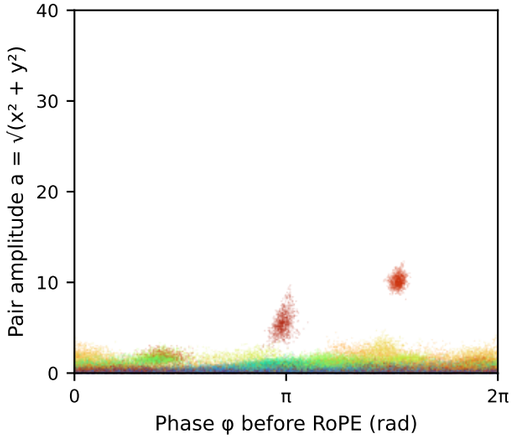
L2 H0



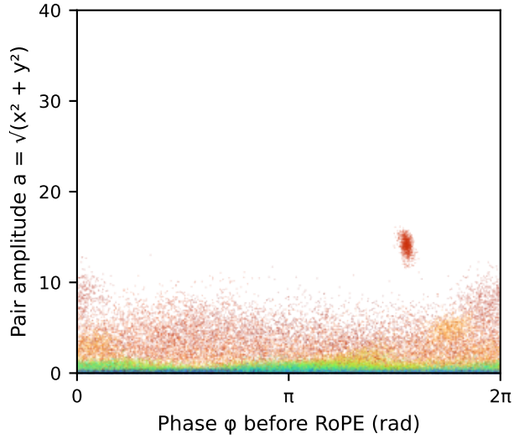
L2 H1



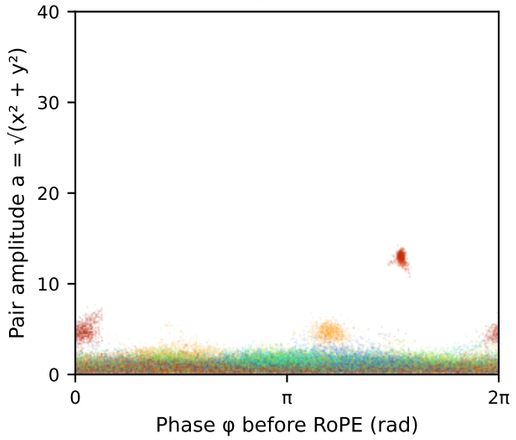
L2 H2



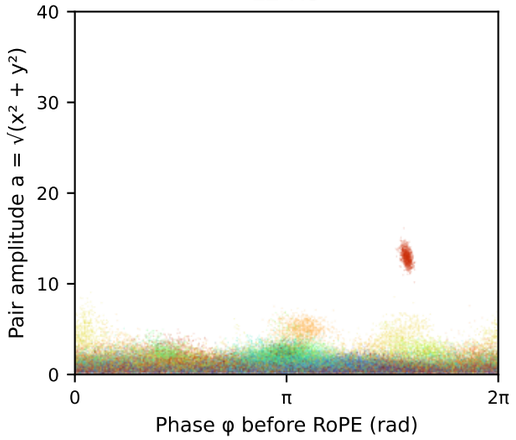
L2 H3



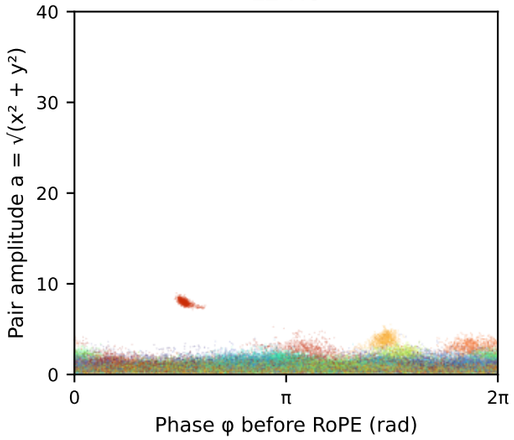
L2 H4



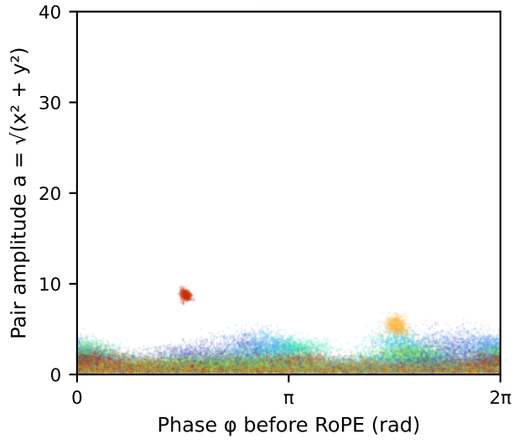
L2 H5



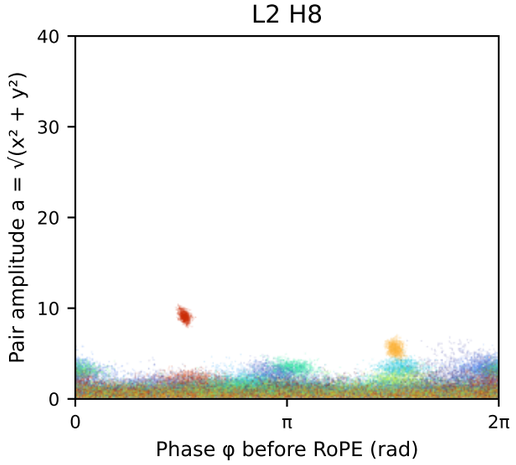
L2 H6



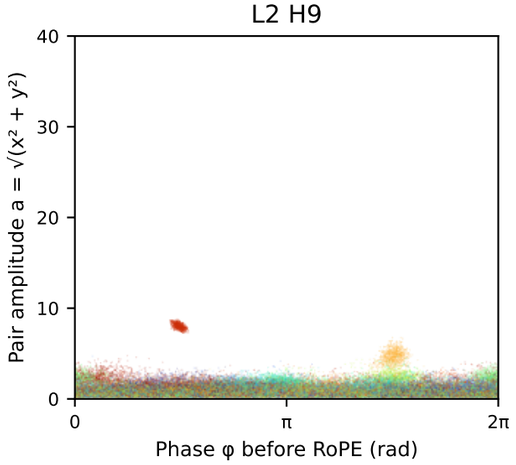
L2 H7



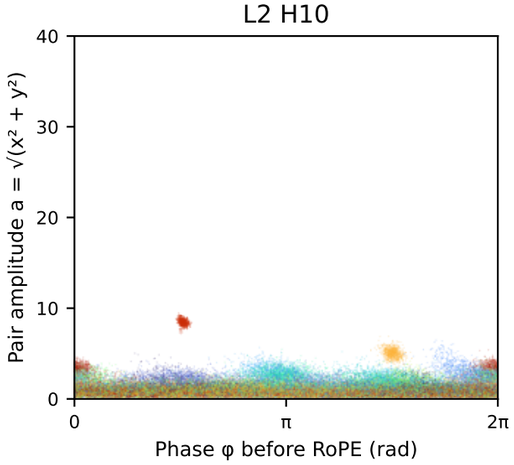
L2 H8



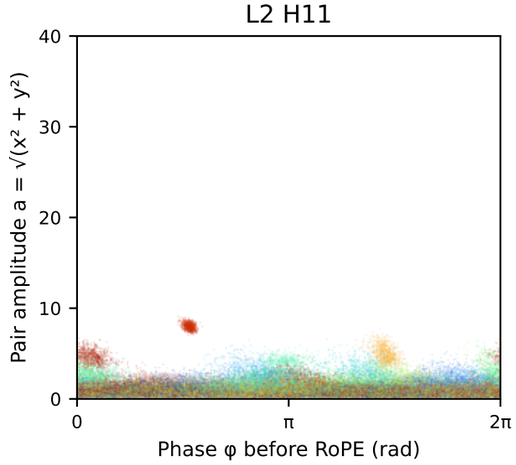
L2 H9

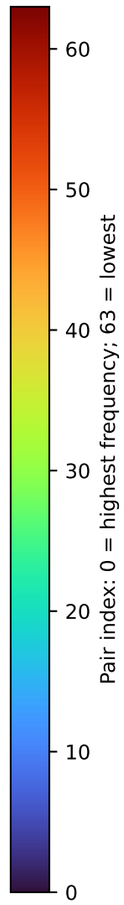


L2 H10



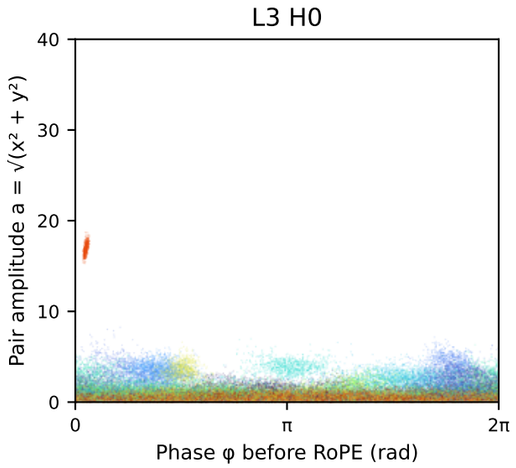
L2 H11

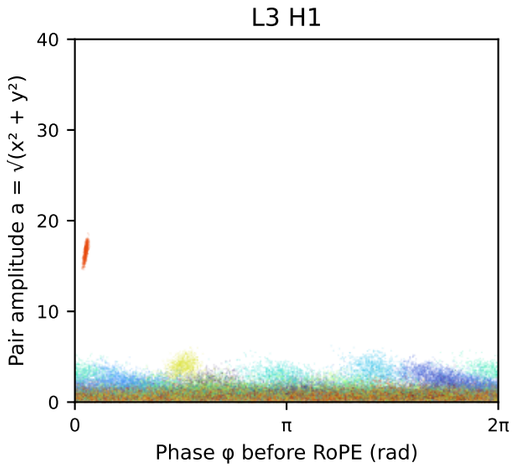


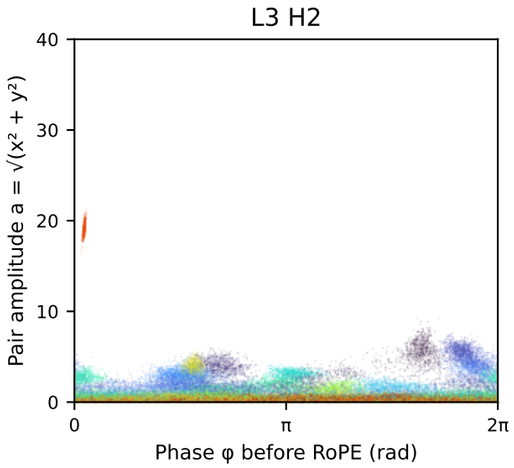


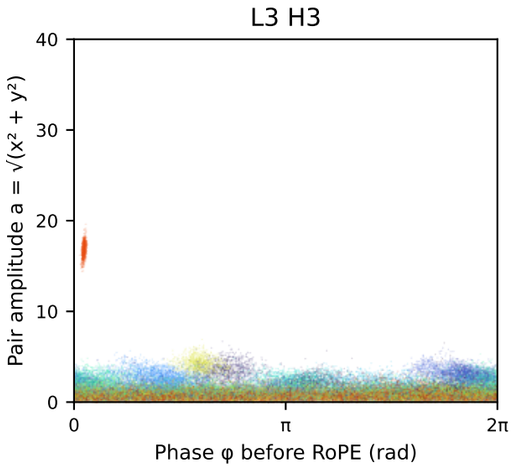
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

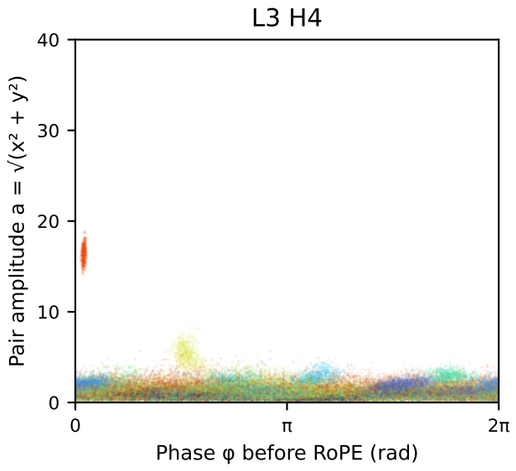
Qwen queries · layer 3 · amplitude axis 0–40

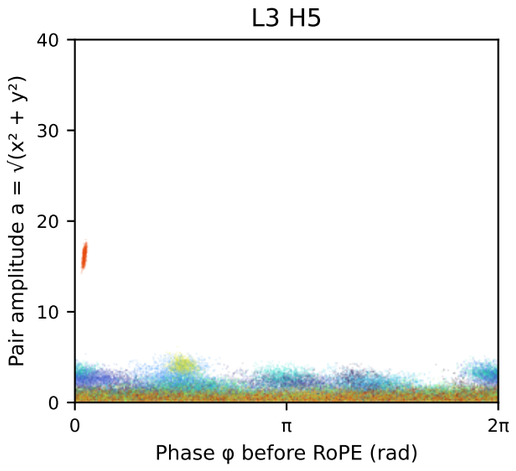


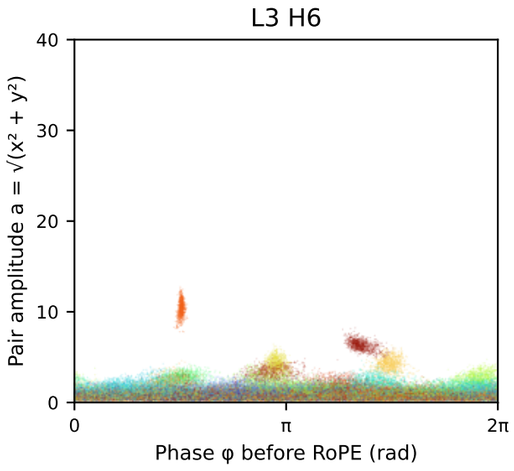


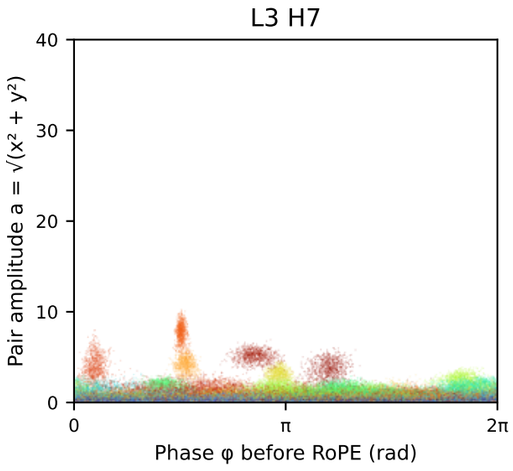


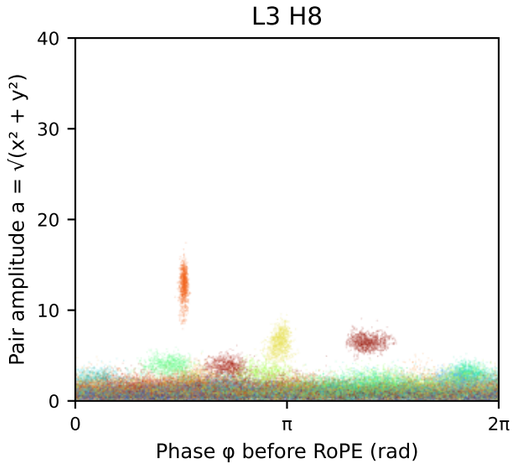


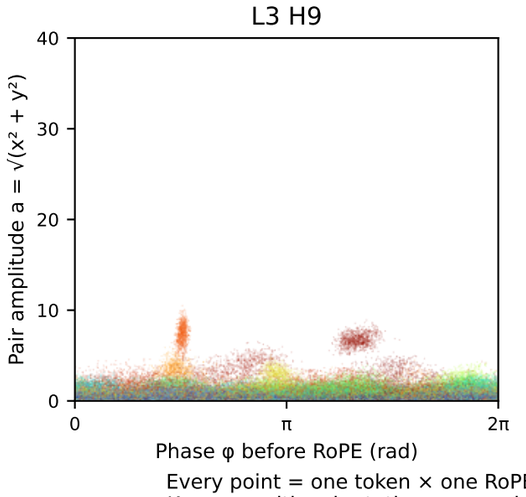


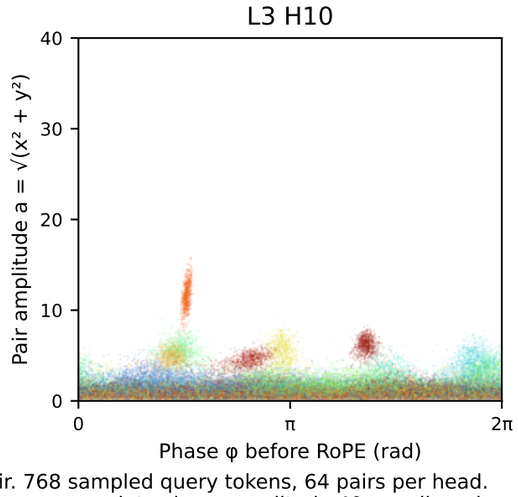


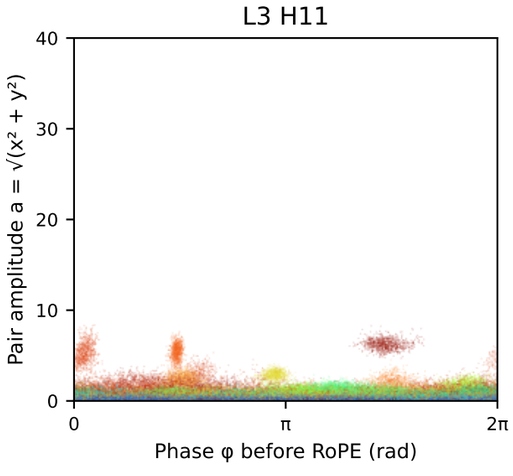


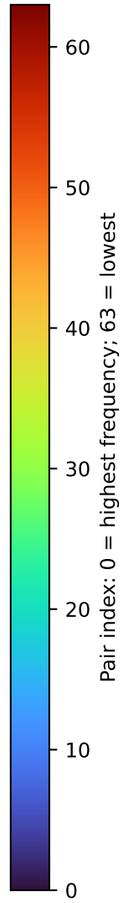






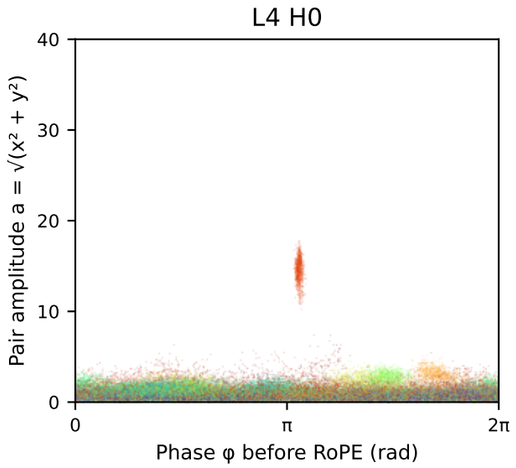


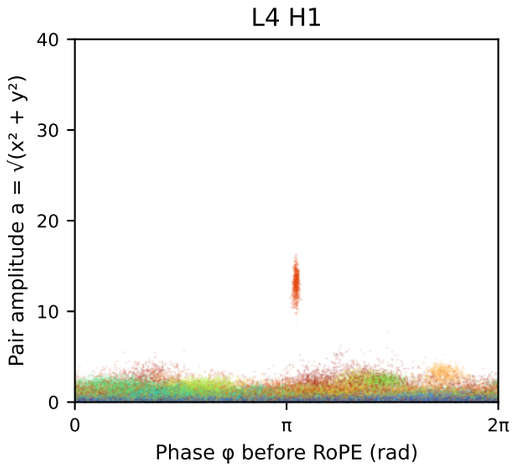


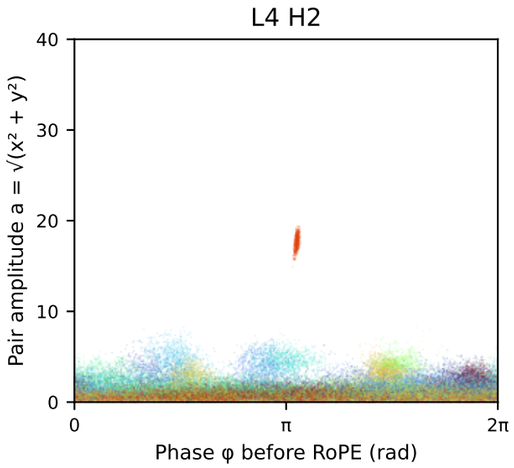


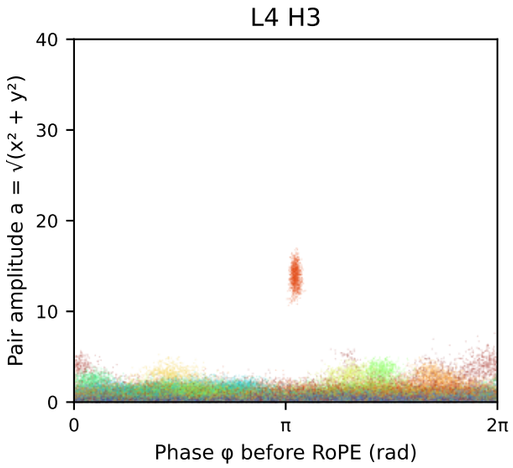
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

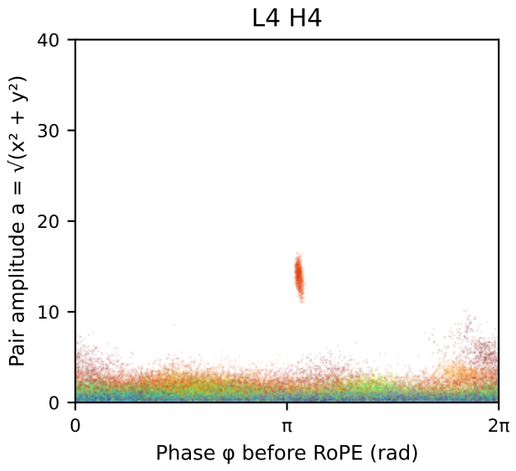
Qwen queries · layer 4 · amplitude axis 0–40

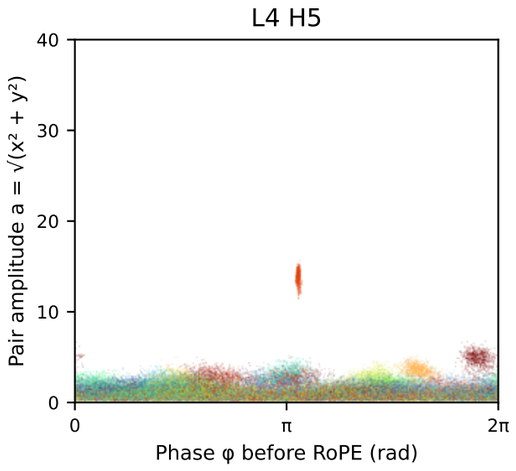


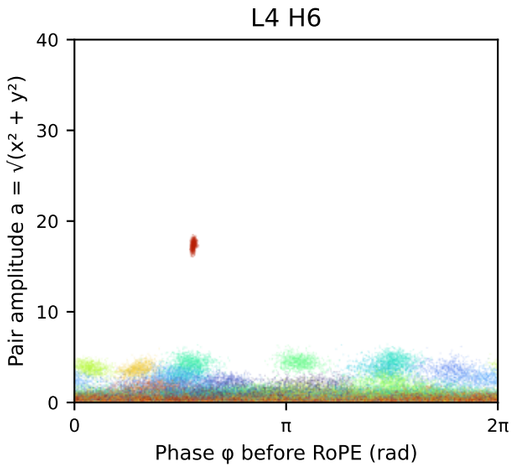


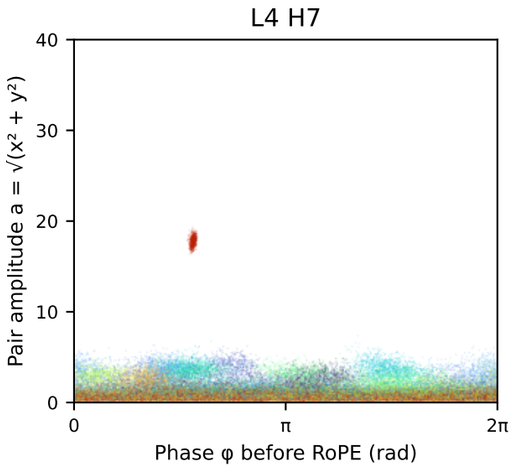


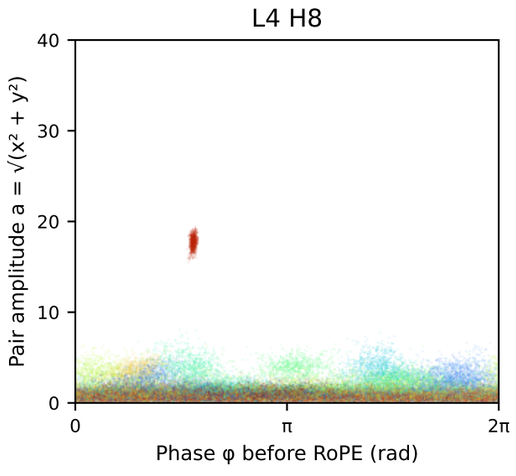


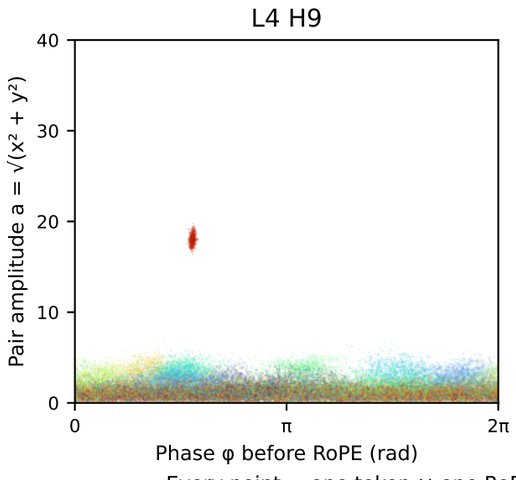


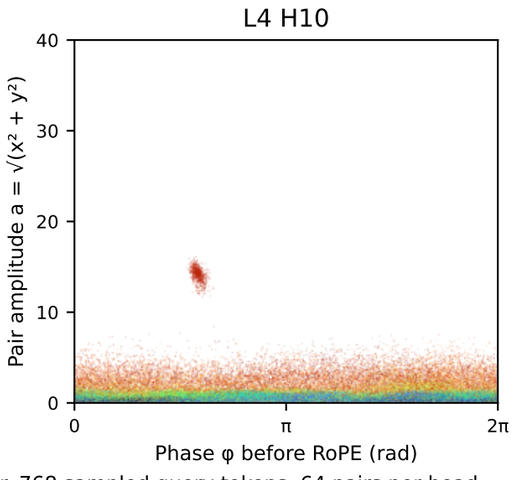


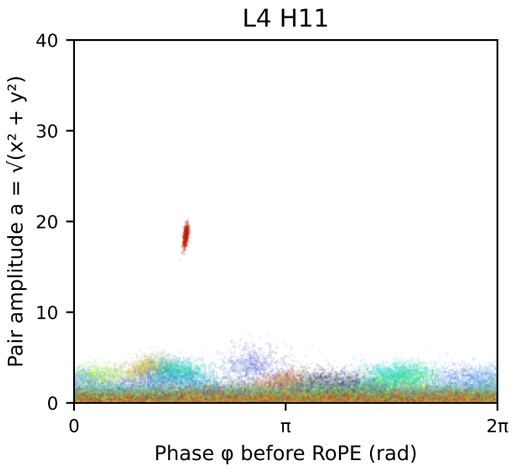


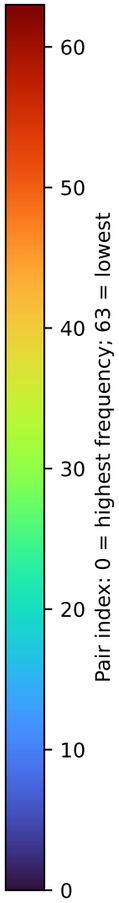






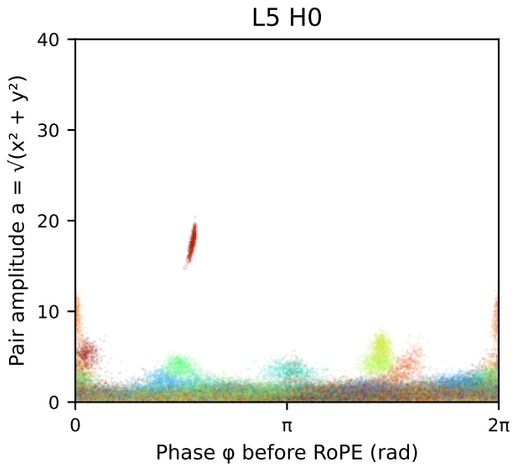


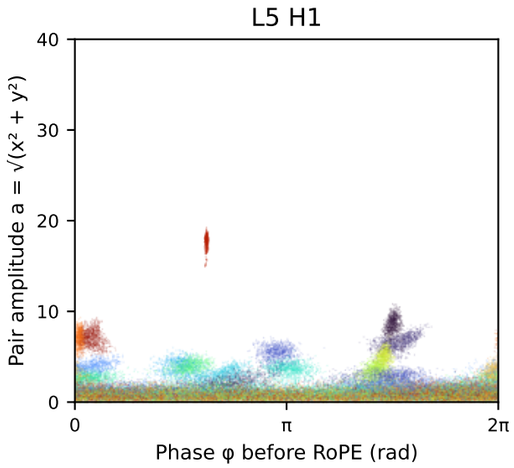


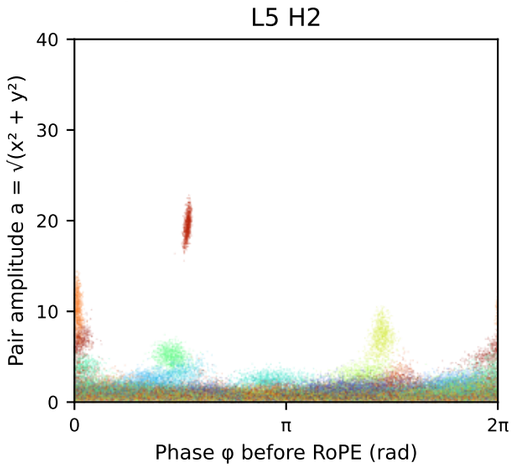


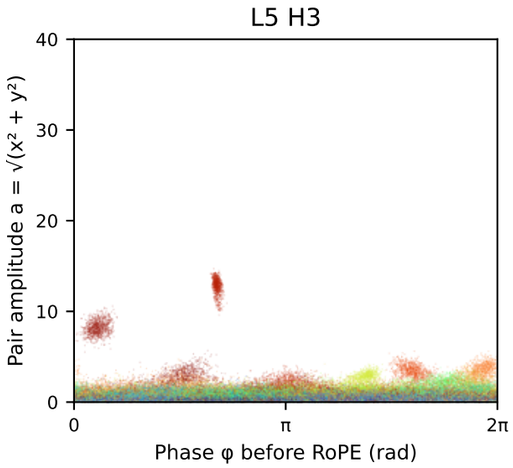
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

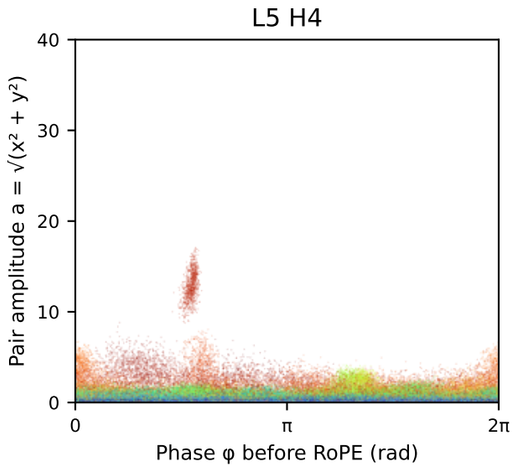
Qwen queries · layer 5 · amplitude axis 0–40

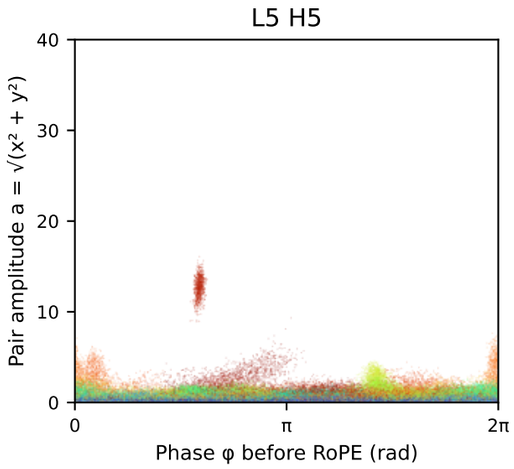


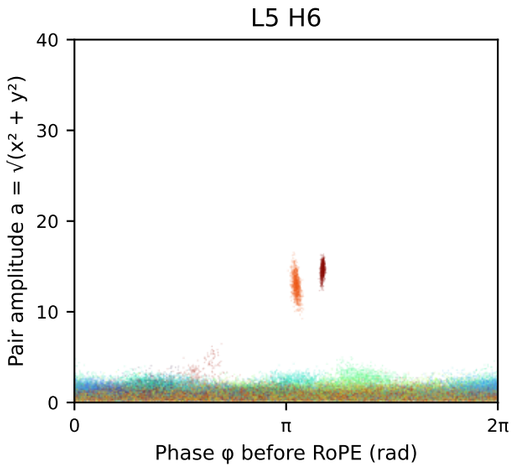


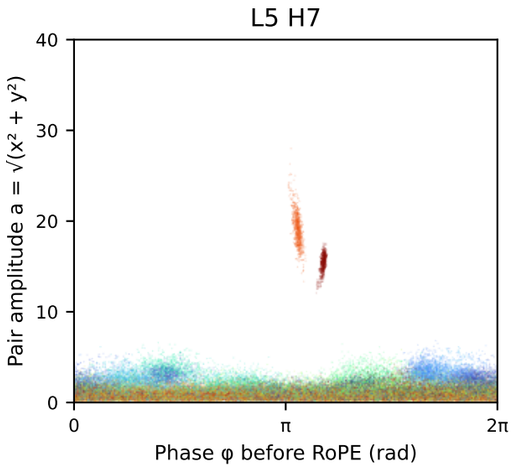


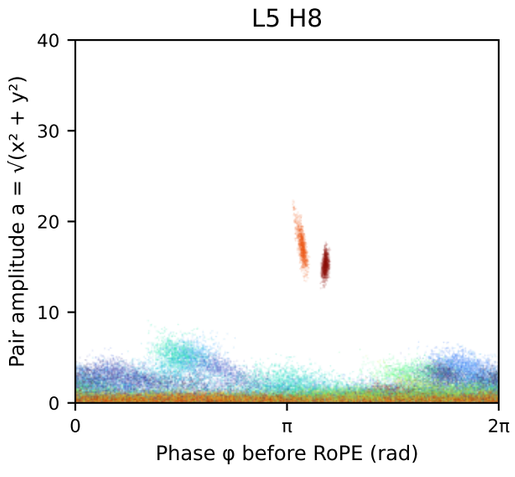


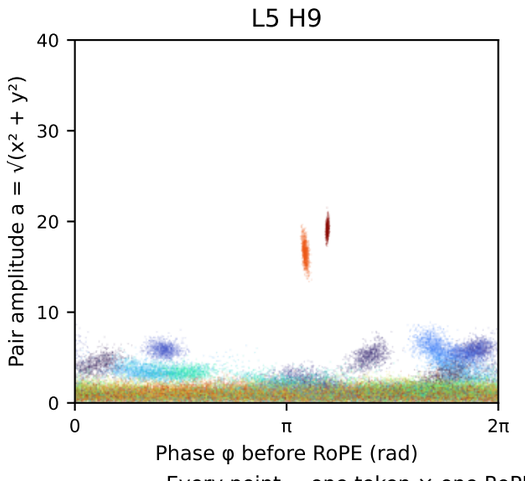


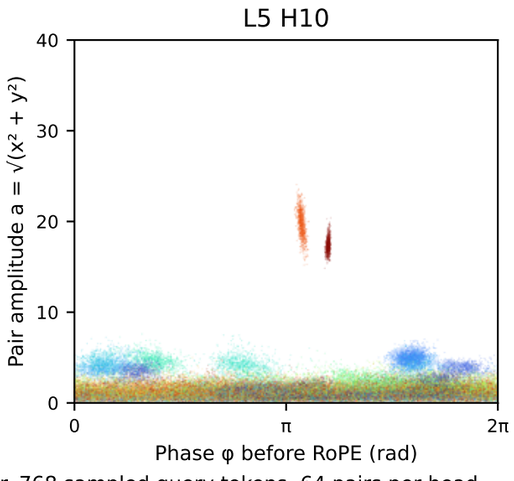


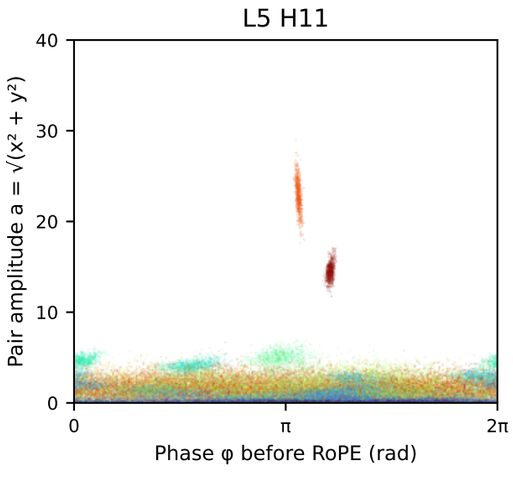


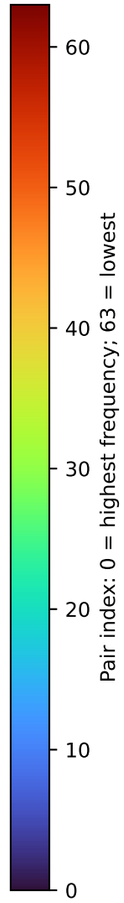








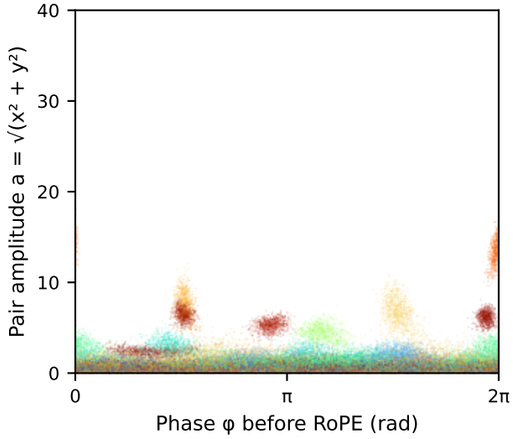




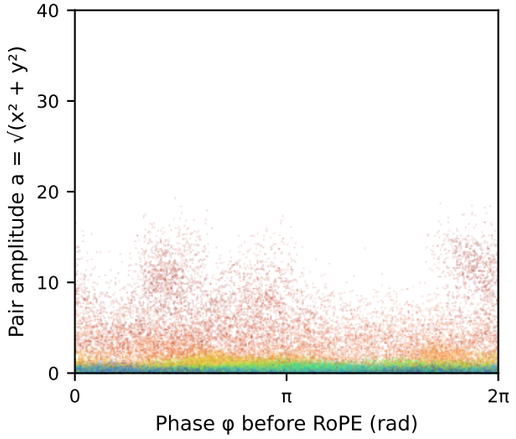
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 6 · amplitude axis 0–40

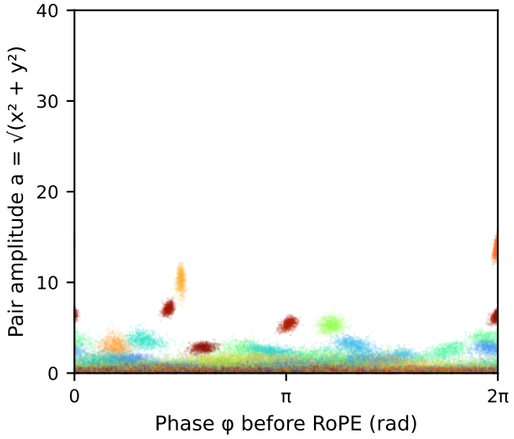
L6 H0



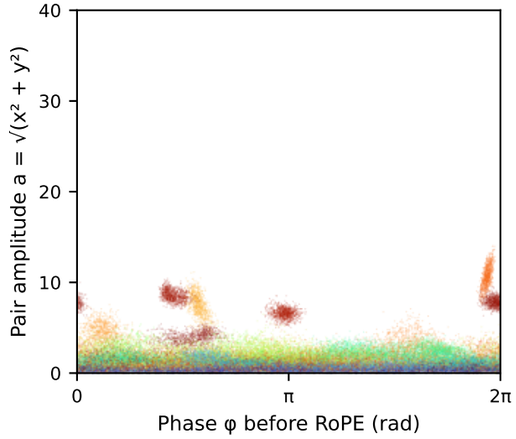
L6 H1



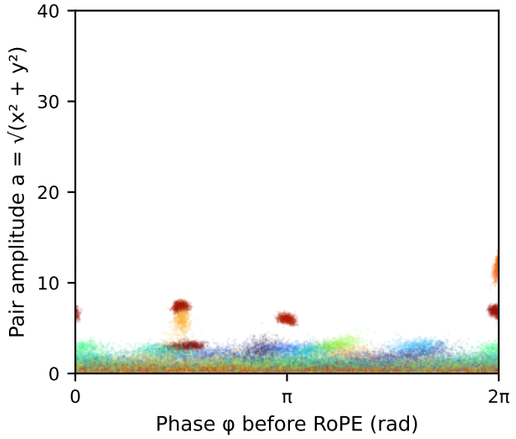
L6 H2



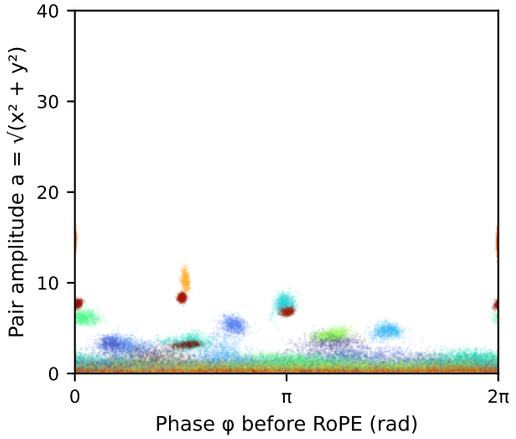
L6 H3



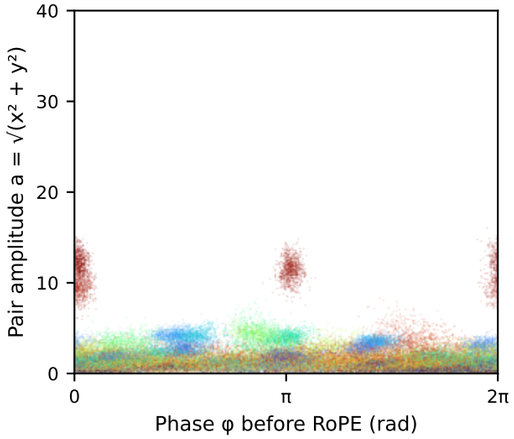
L6 H4



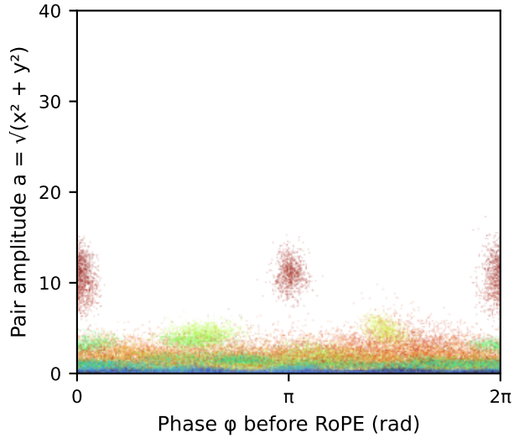
L6 H5



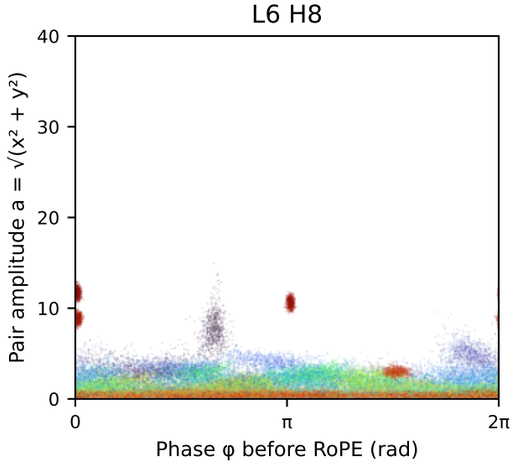
L6 H6



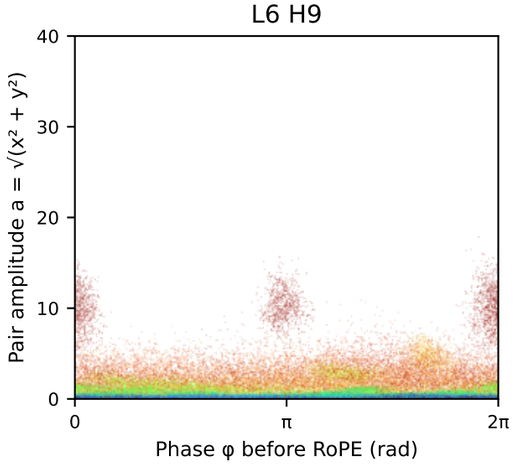
L6 H7



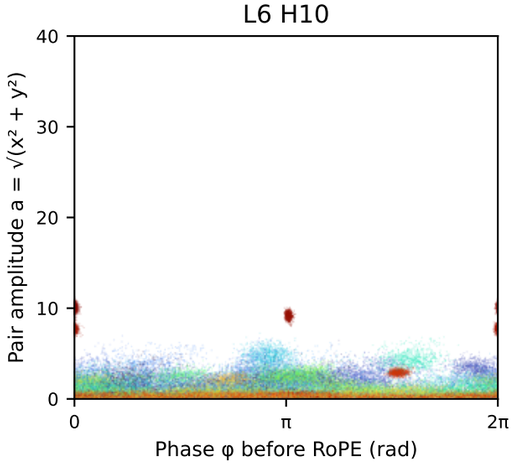
L6 H8



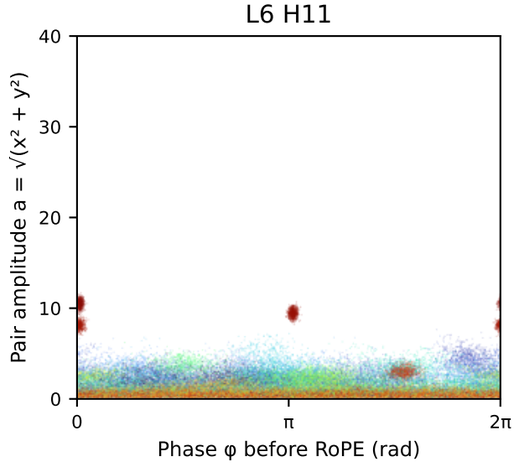
L6 H9

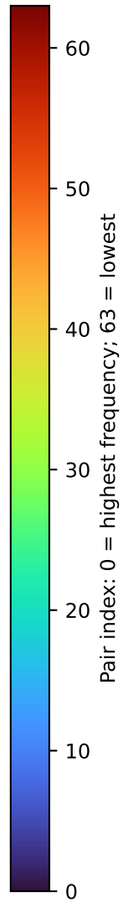


L6 H10



L6 H11

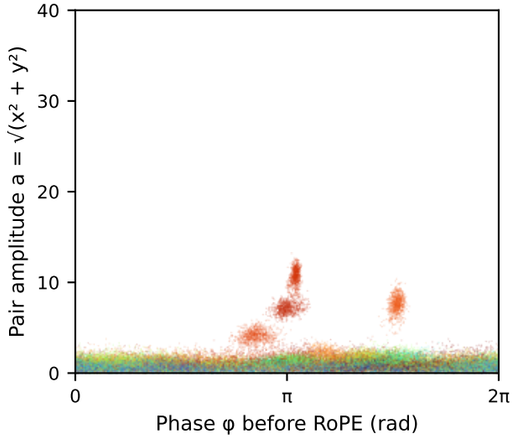




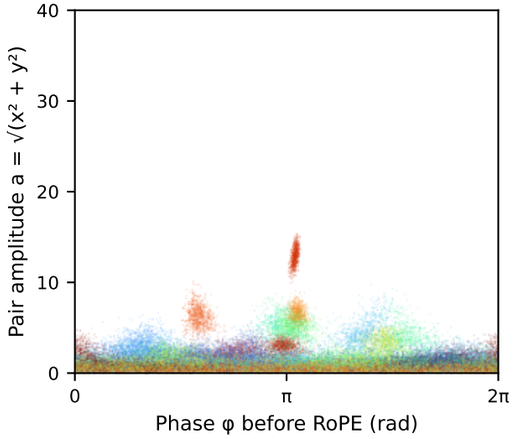
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 7 · amplitude axis 0–40

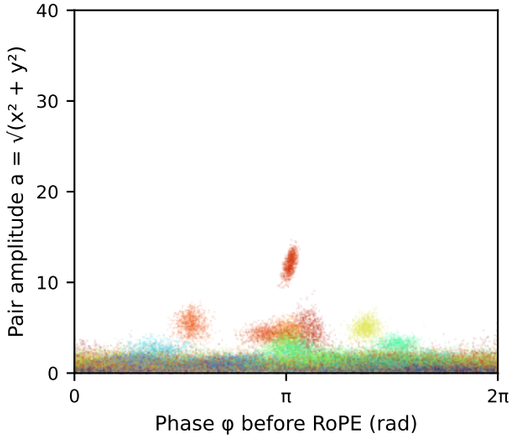
L7 H0



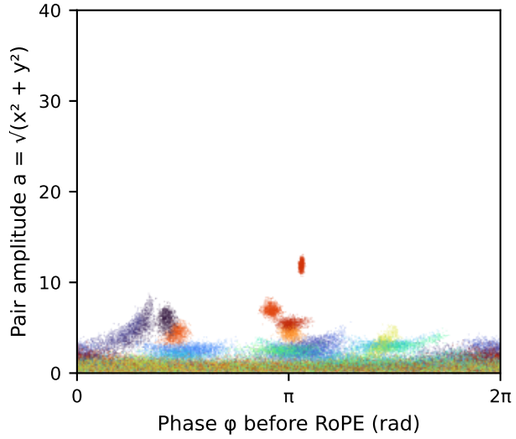
L7 H1



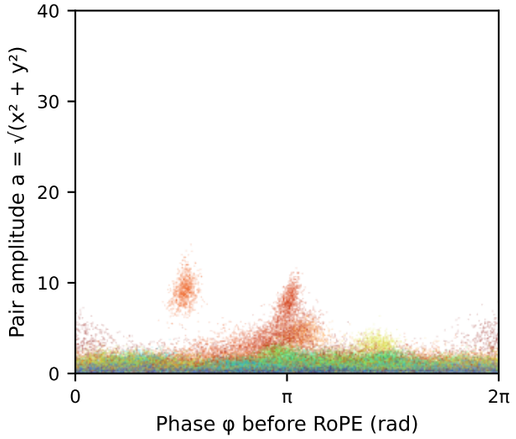
L7 H2



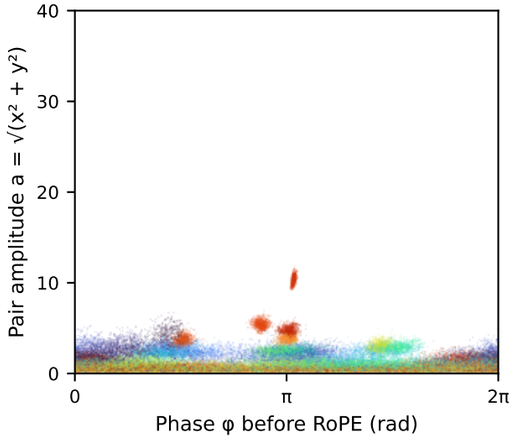
L7 H3



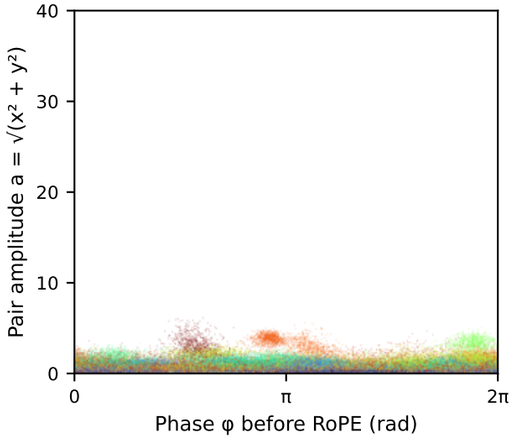
L7 H4



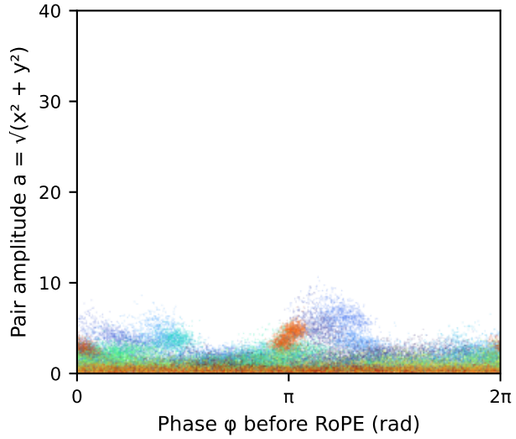
L7 H5



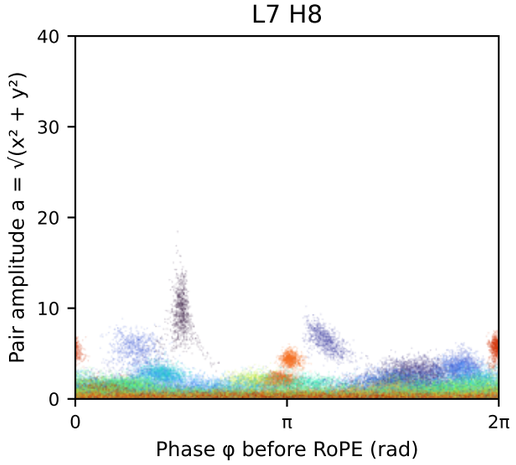
L7 H6



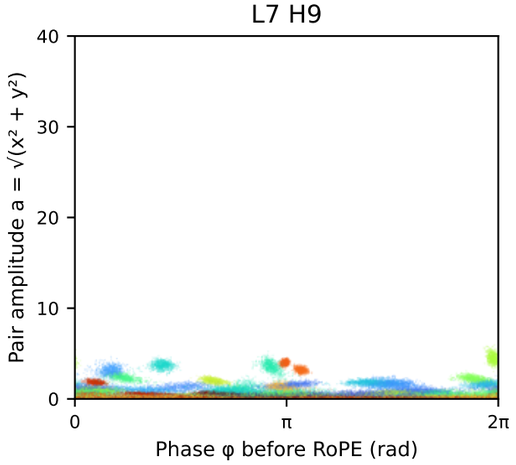
L7 H7



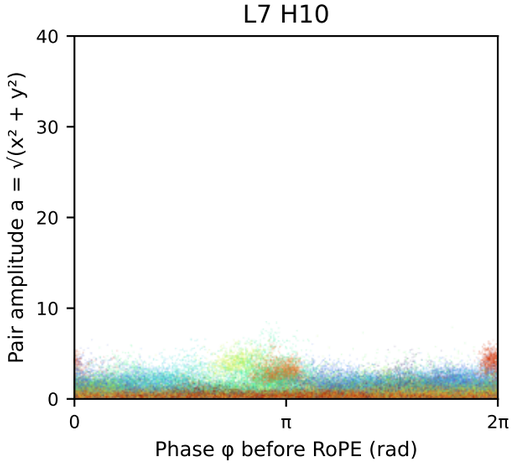
L7 H8



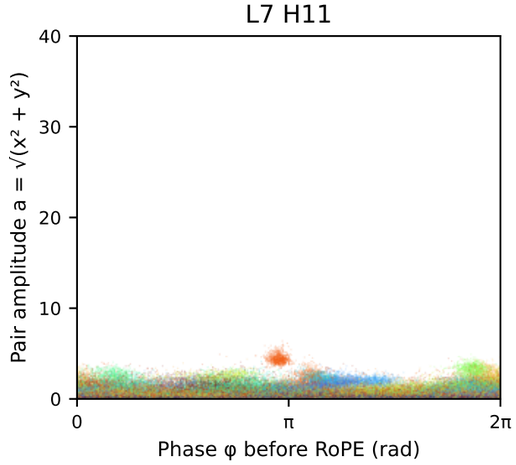
L7 H9

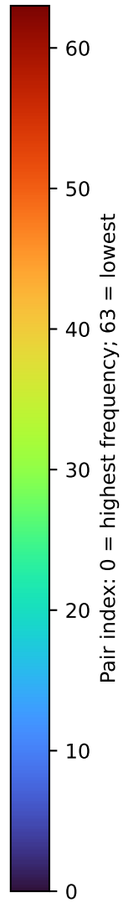


L7 H10



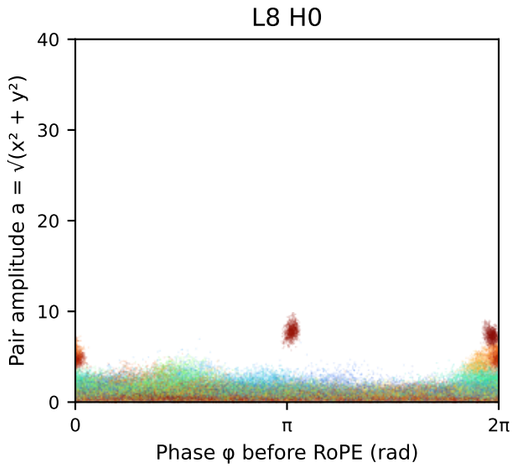
L7 H11

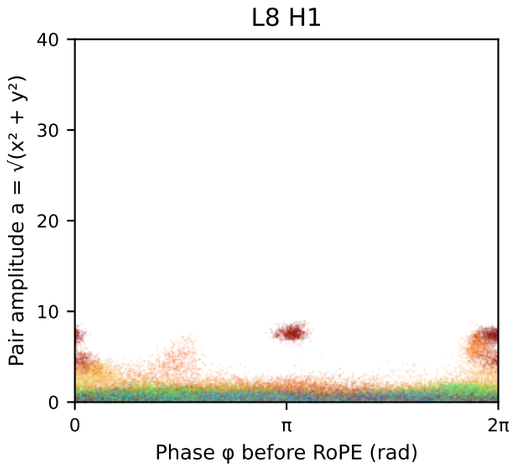


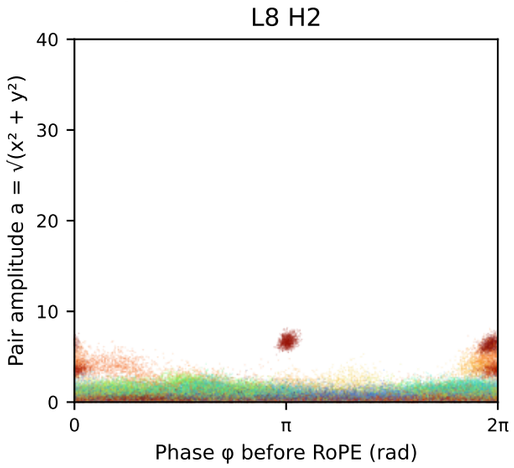


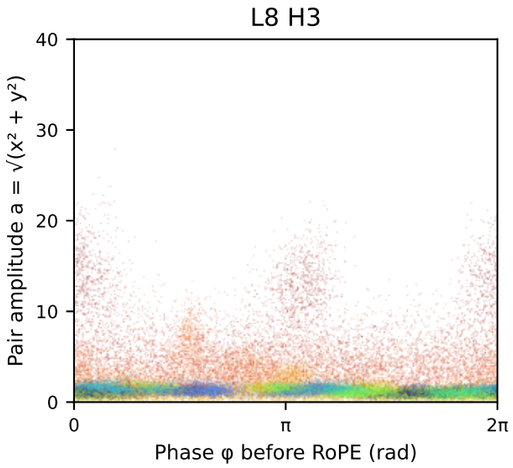
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

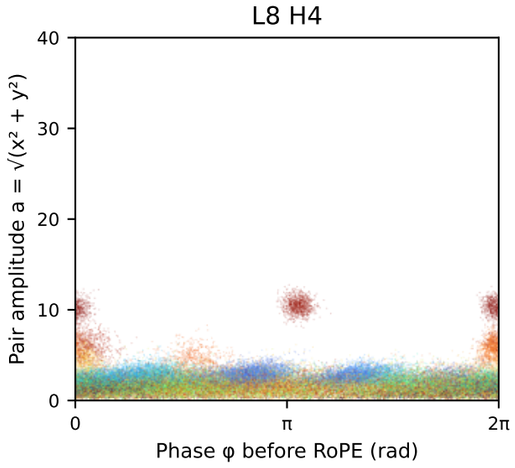
Qwen queries · layer 8 · amplitude axis 0–40

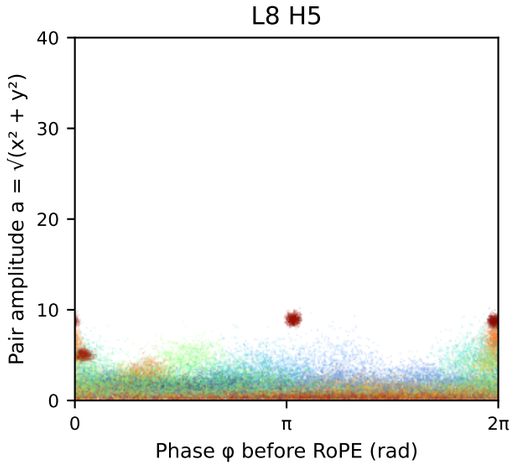


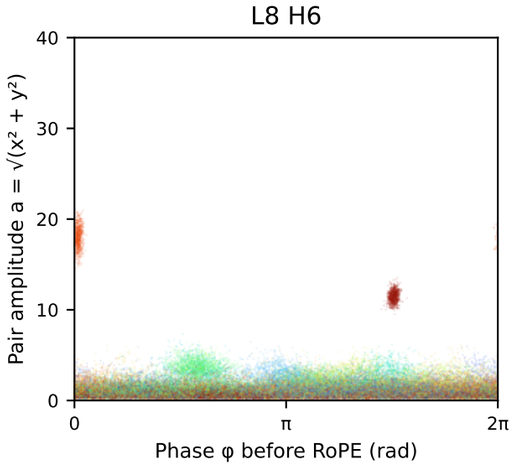


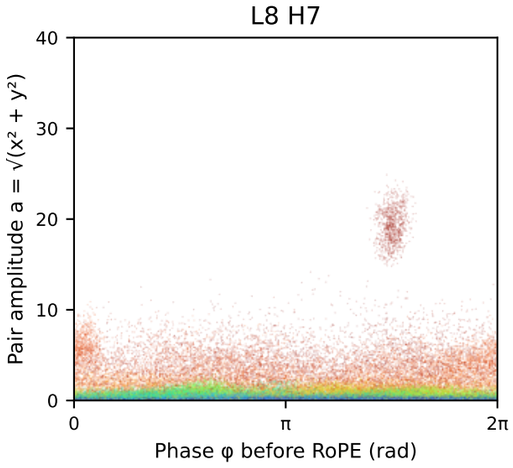


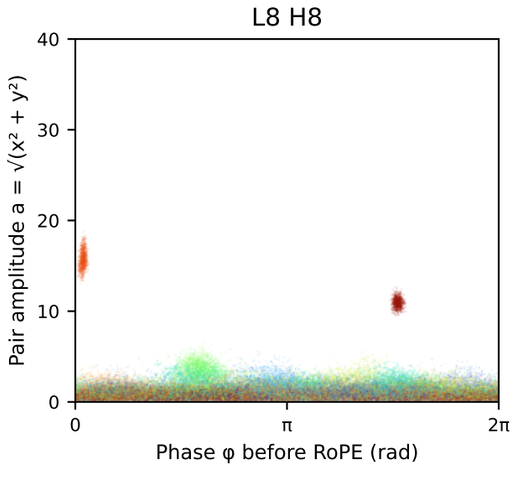


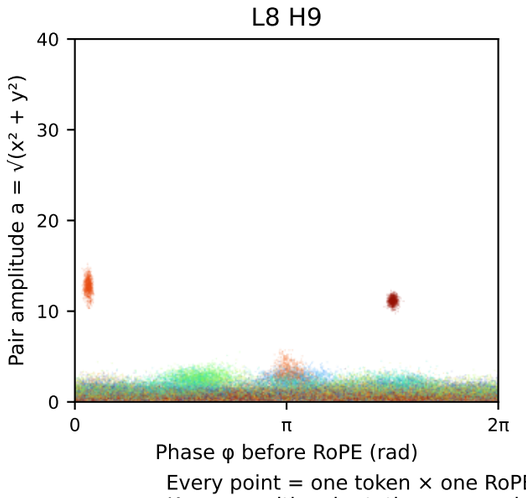


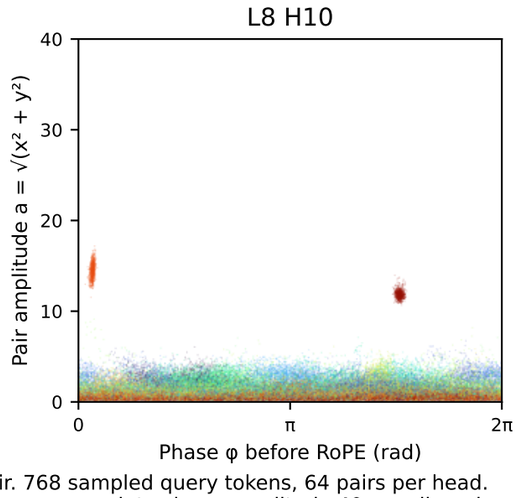


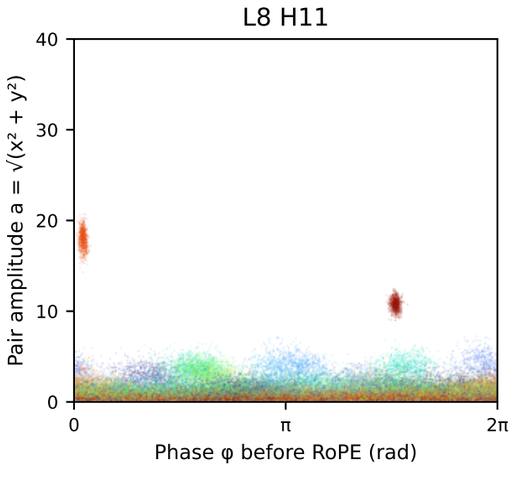


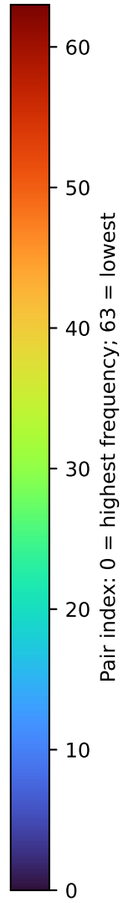








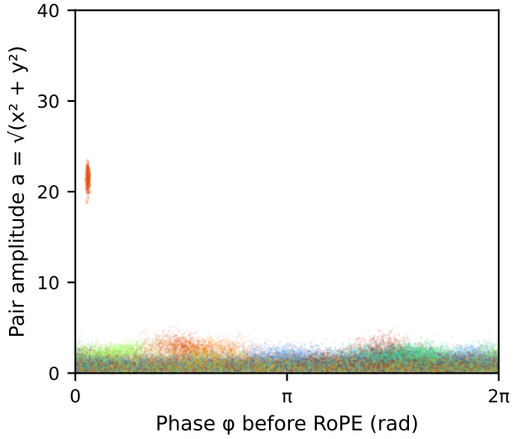




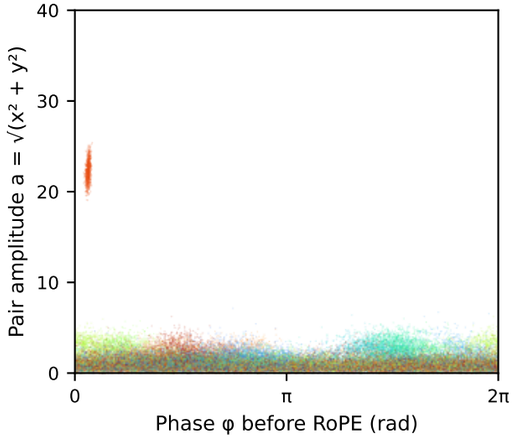
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 9 · amplitude axis 0–40

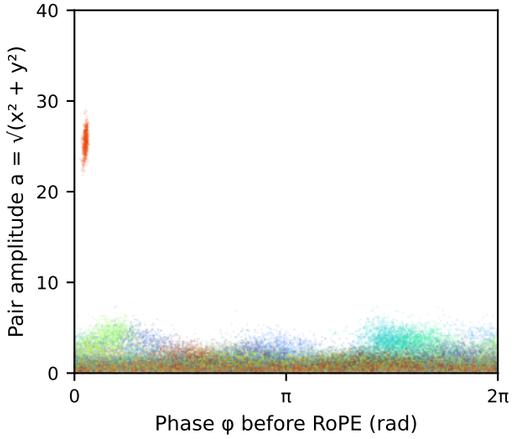
L9 H0



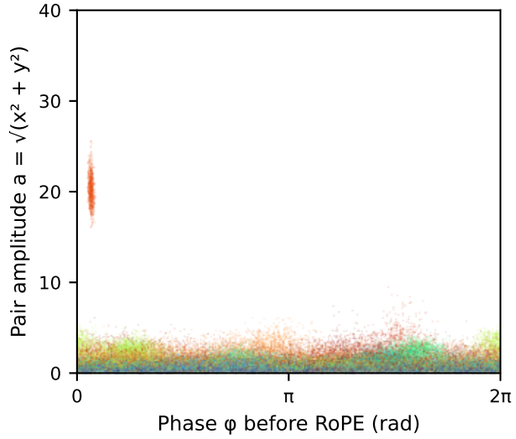
L9 H1



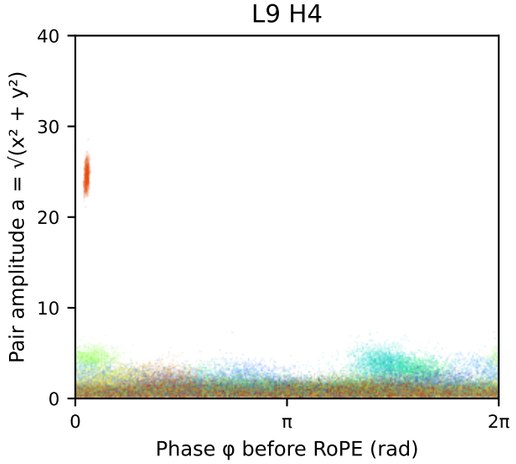
L9 H2



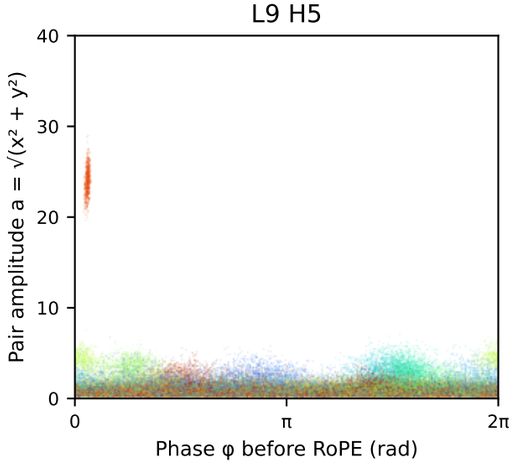
L9 H3



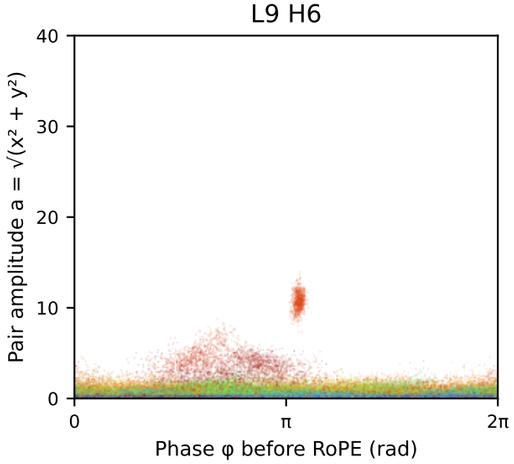
L9 H4



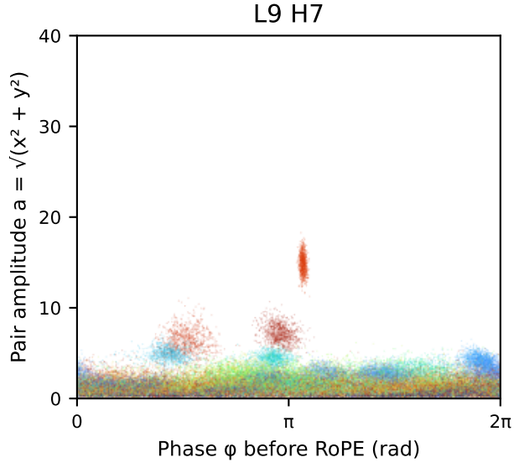
L9 H5



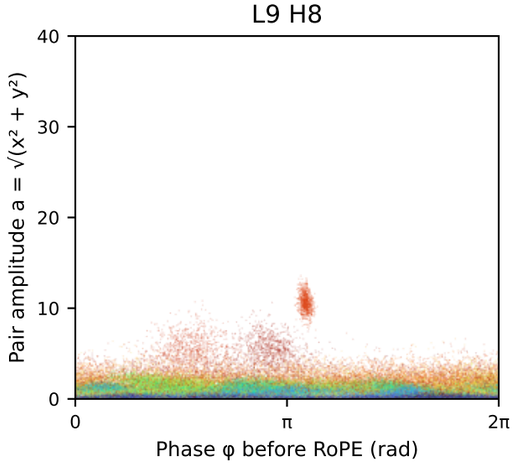
L9 H6



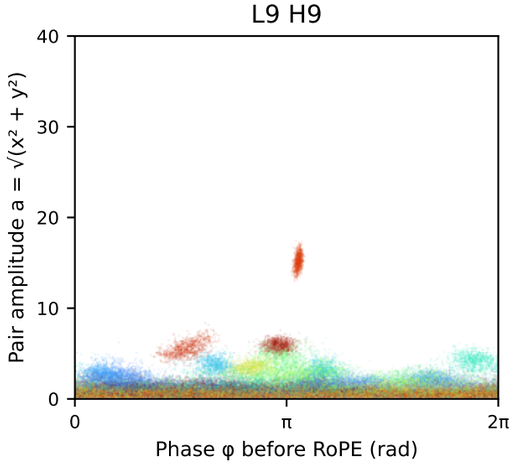
L9 H7



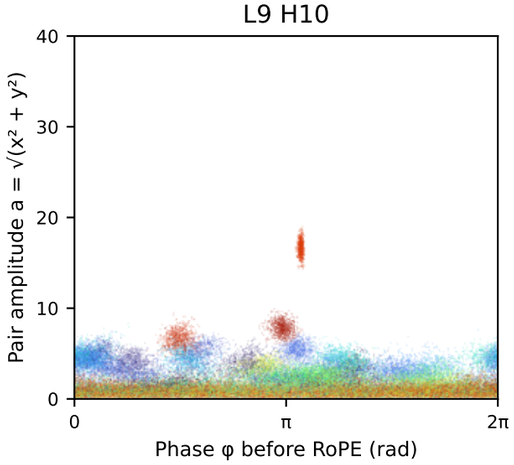
L9 H8



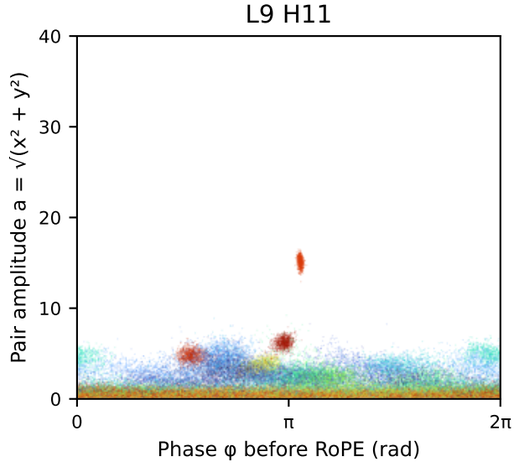
L9 H9

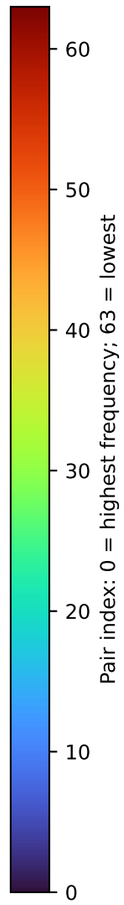


L9 H10



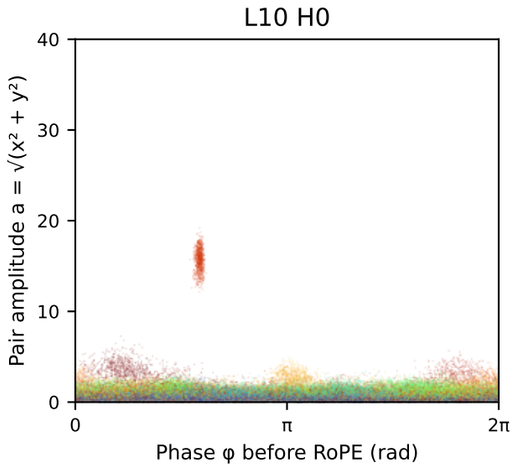
L9 H11

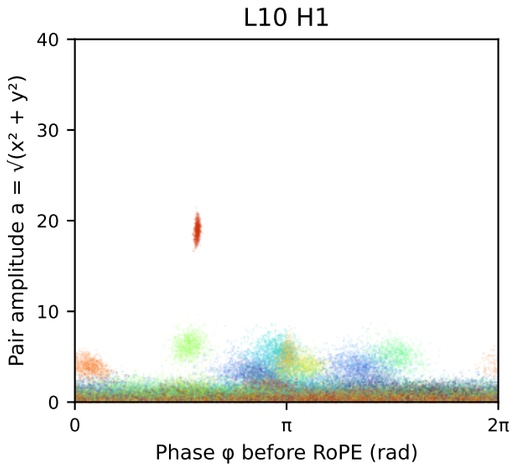


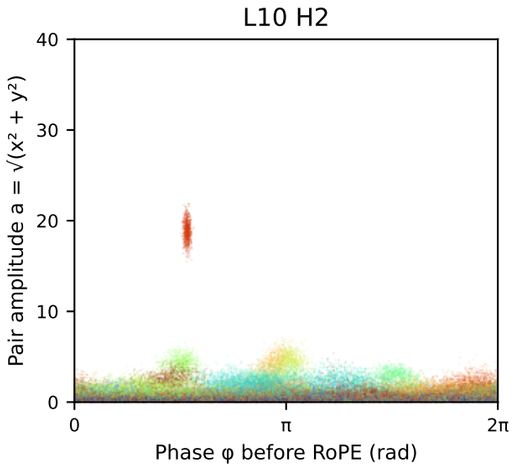


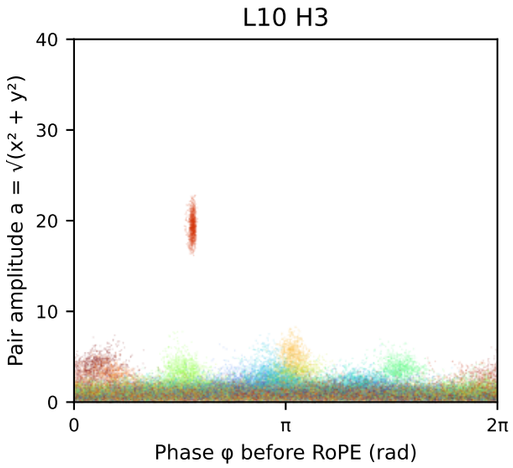
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

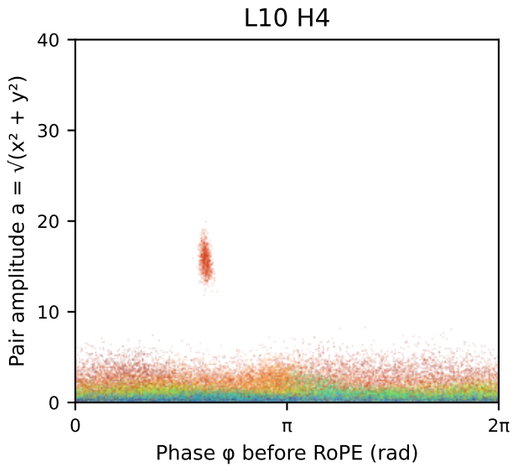
Qwen queries · layer 10 · amplitude axis 0–40

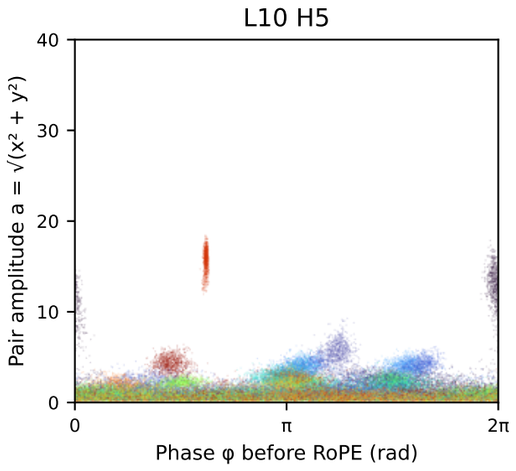


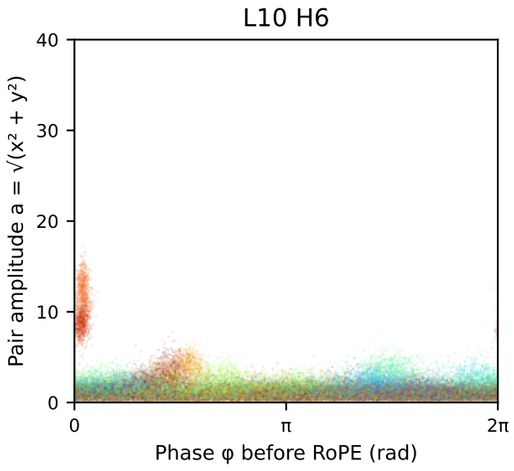


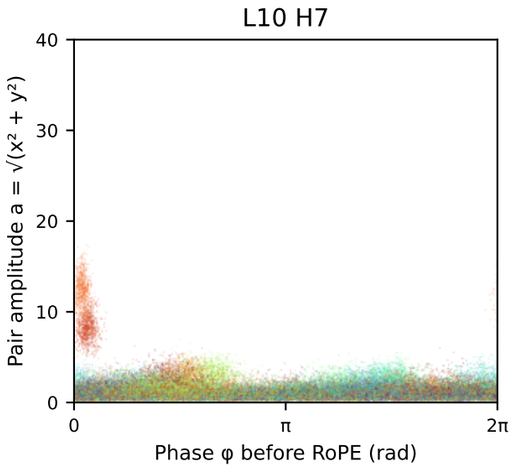


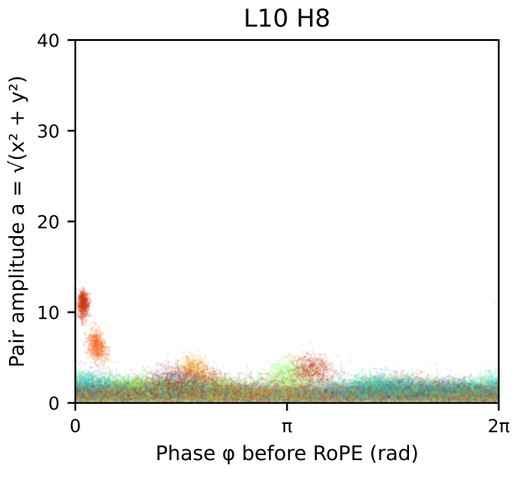


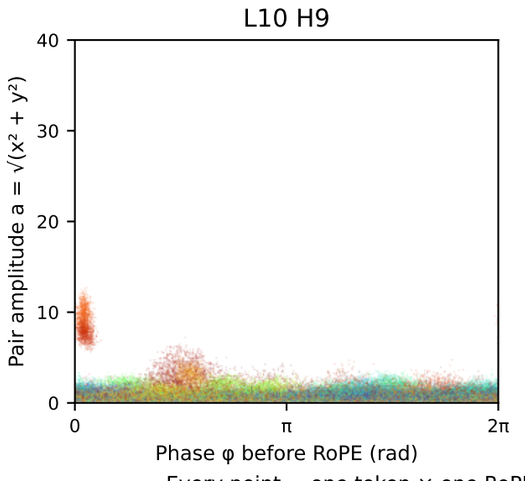


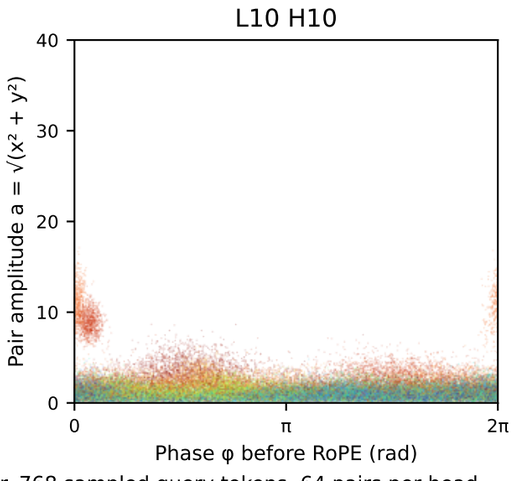


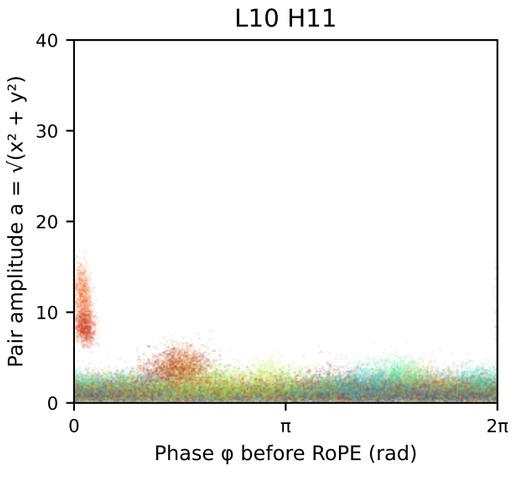


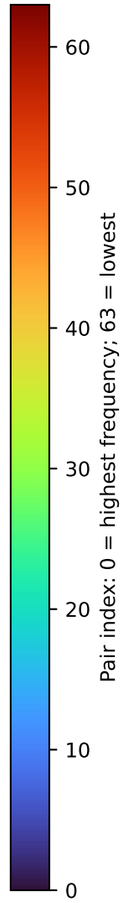






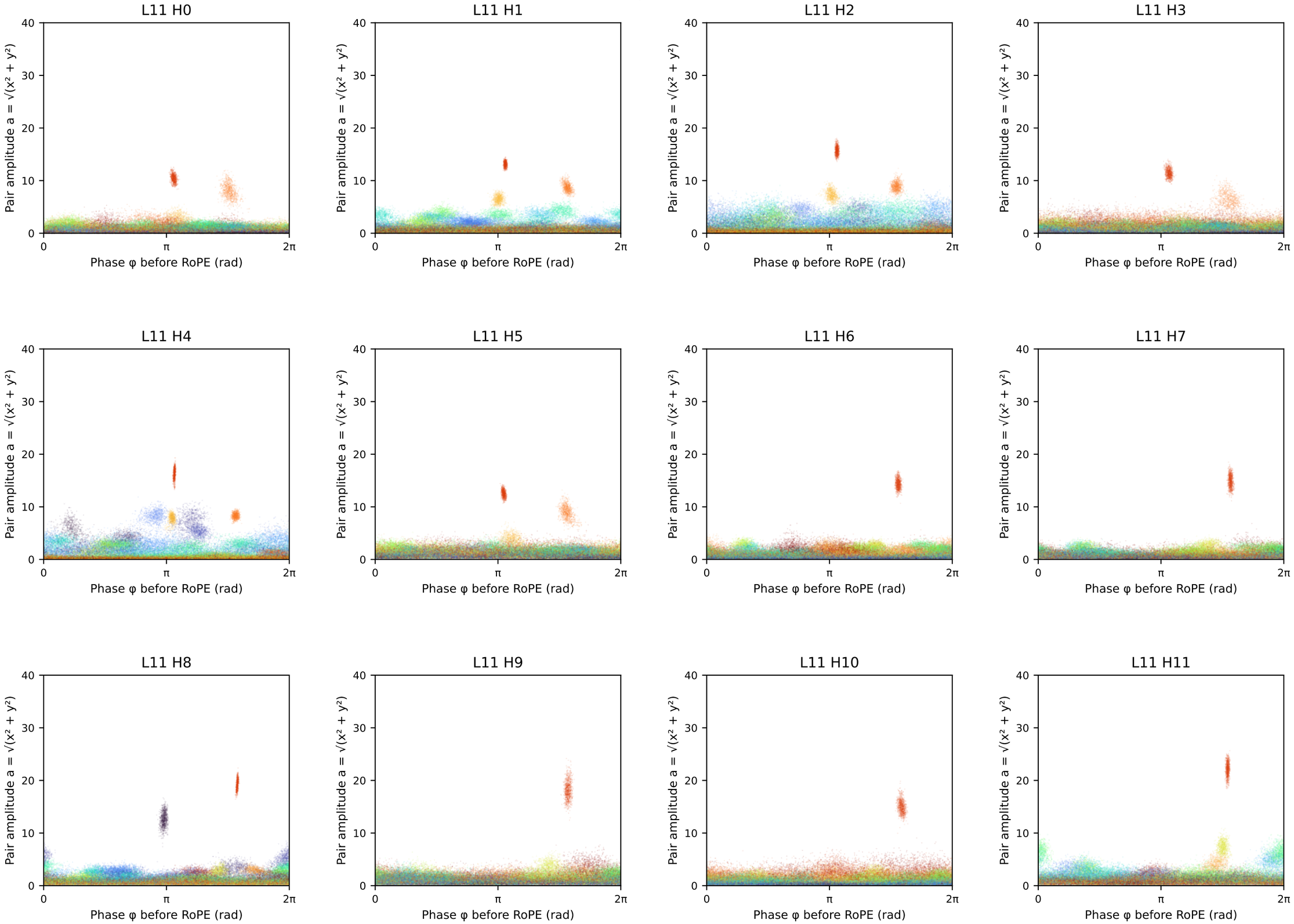






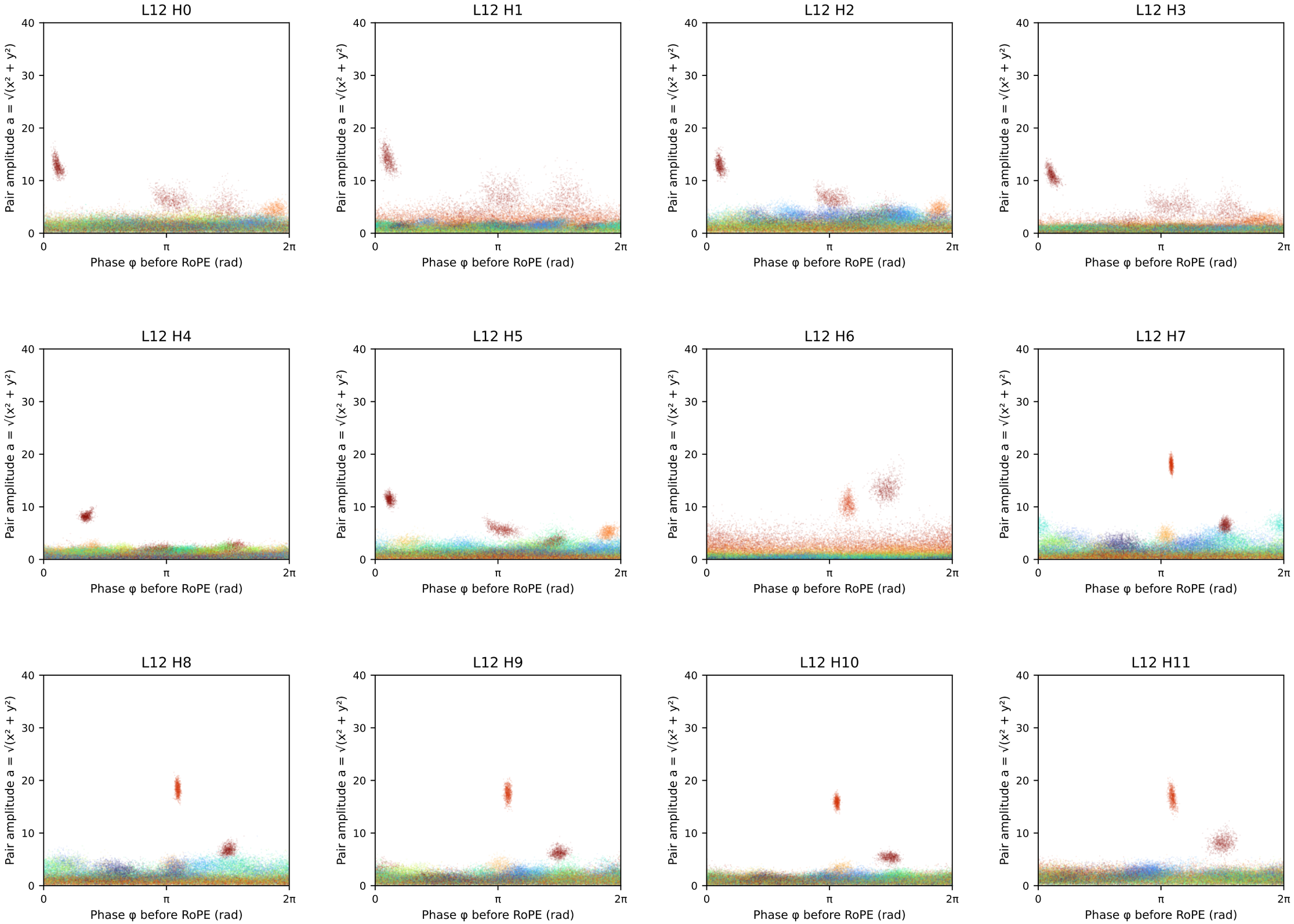
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 11 · amplitude axis 0–40



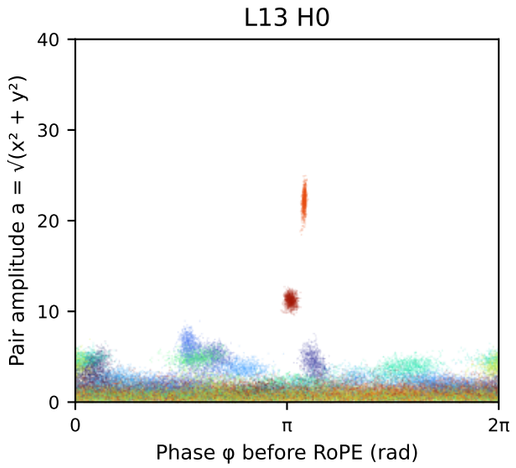
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

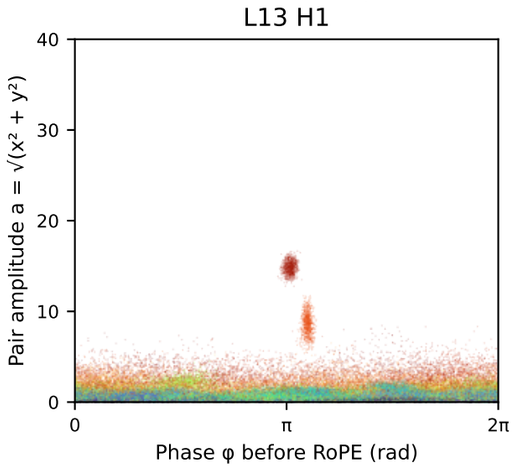
Qwen queries · layer 12 · amplitude axis 0–40

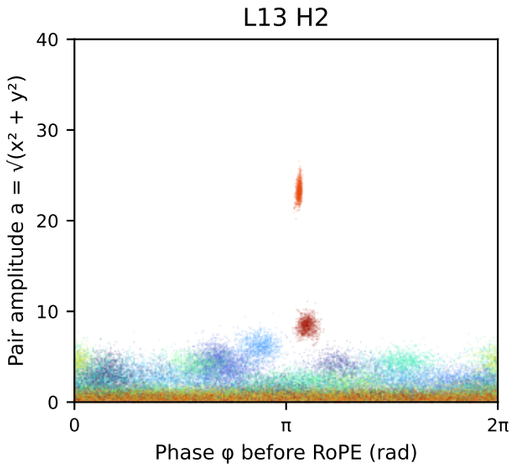


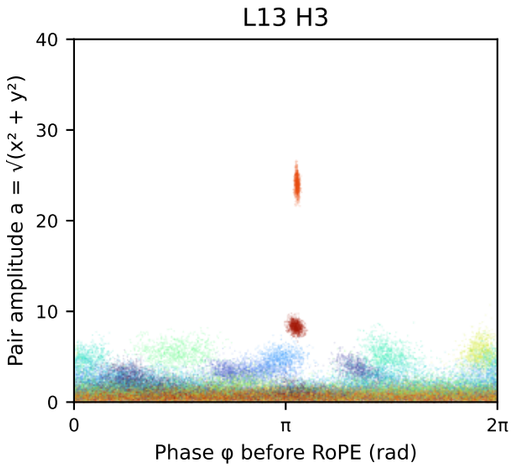
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

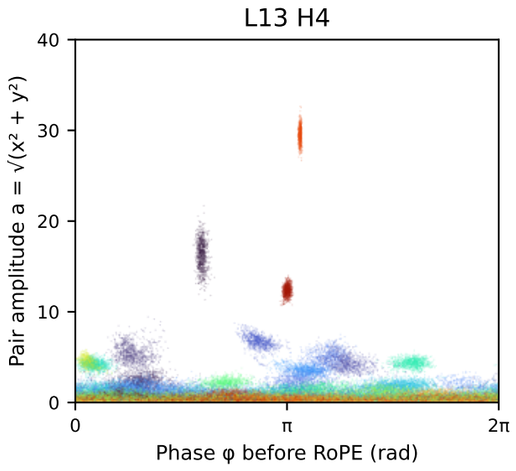
Qwen queries · layer 13 · amplitude axis 0–40

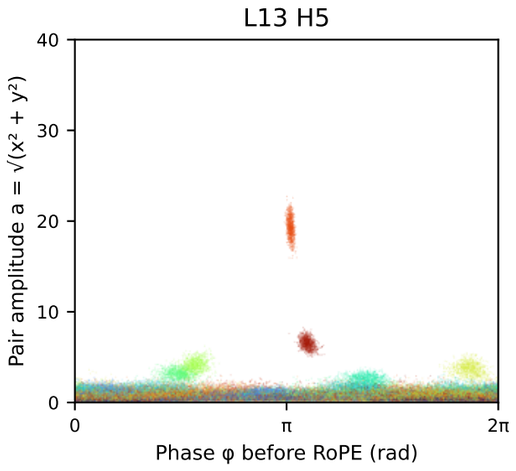


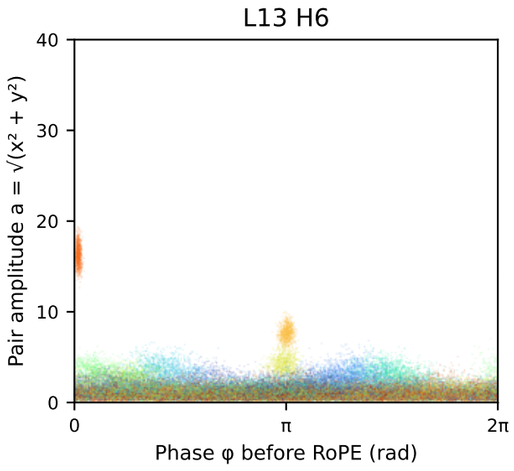


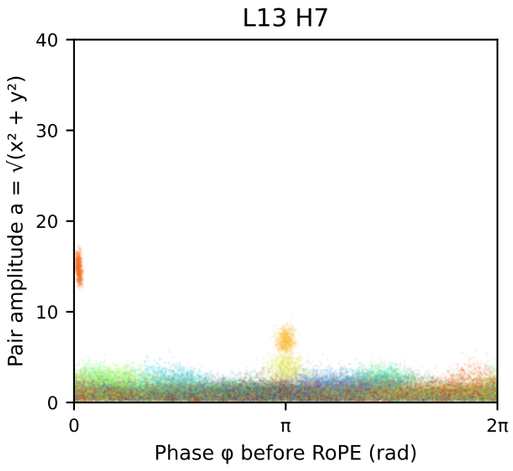


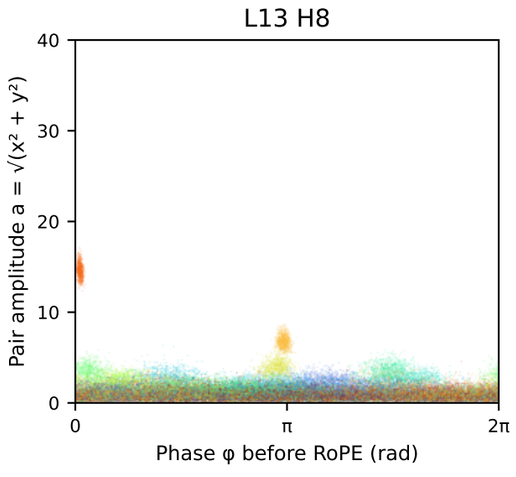


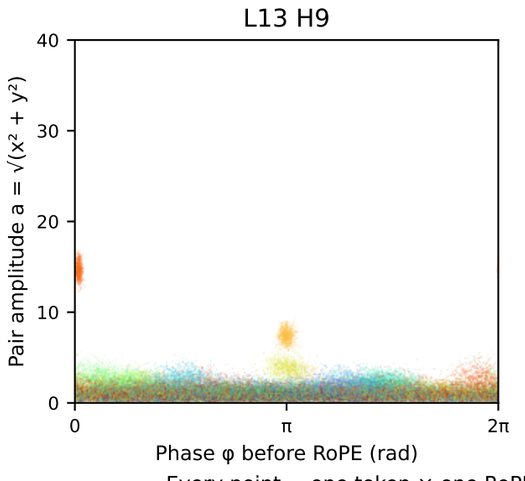


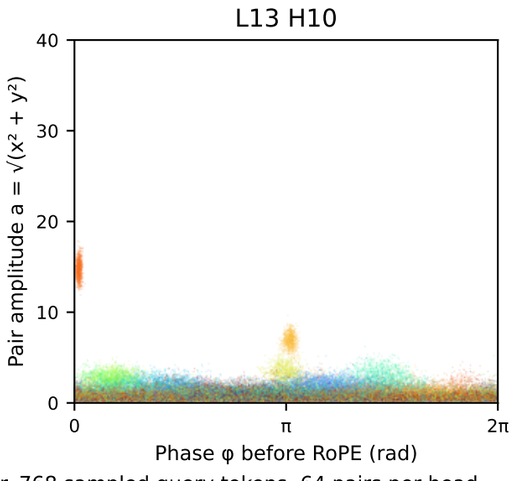


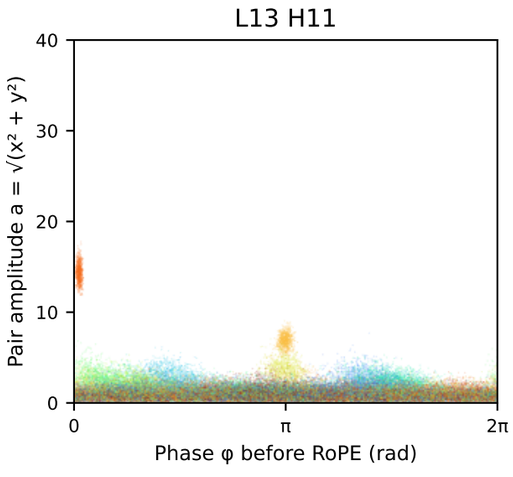


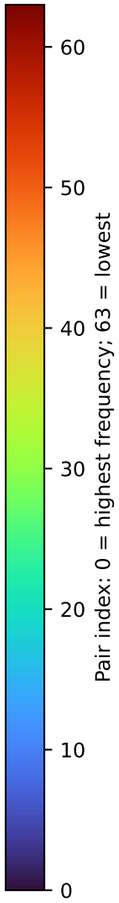






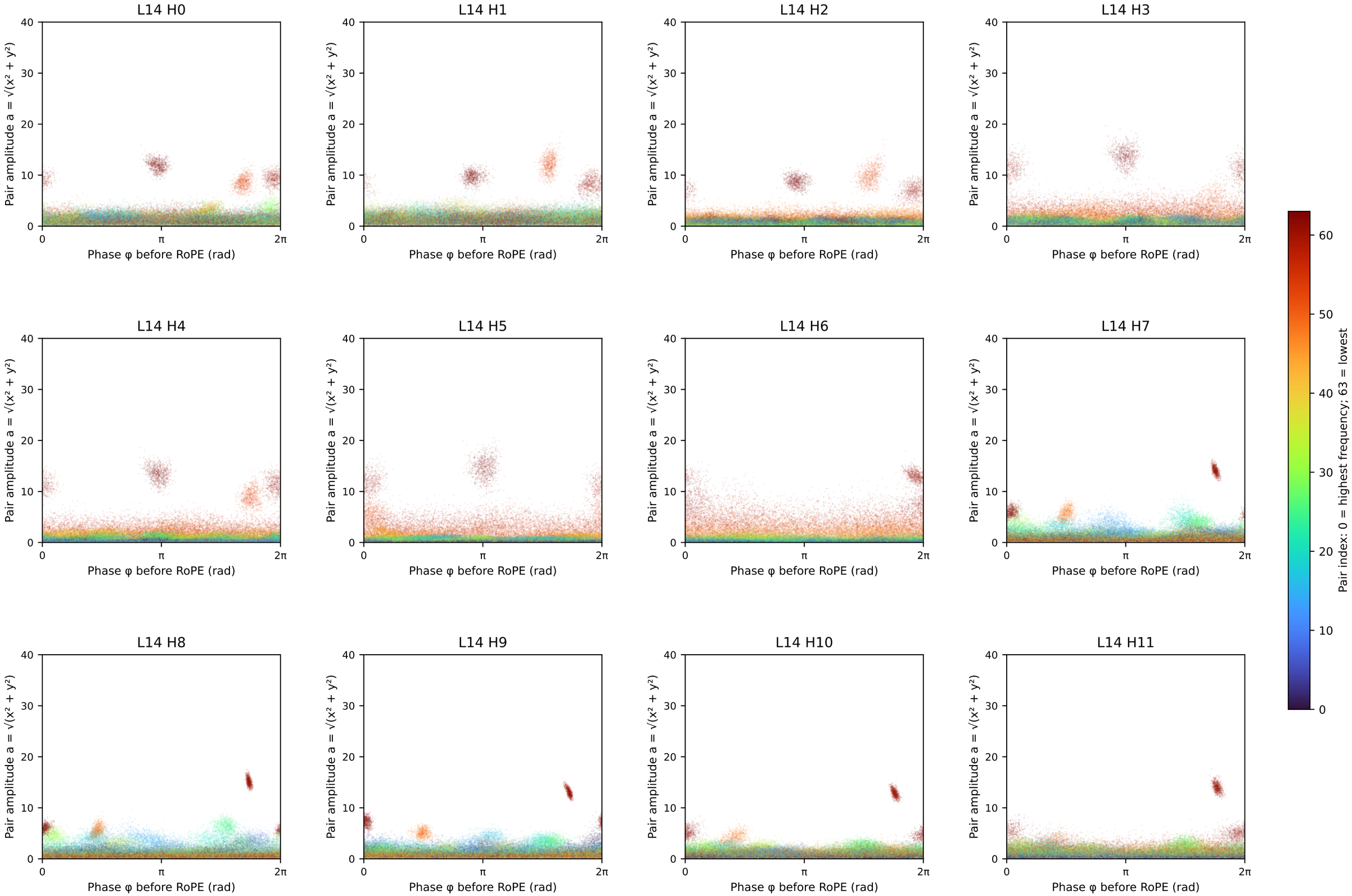






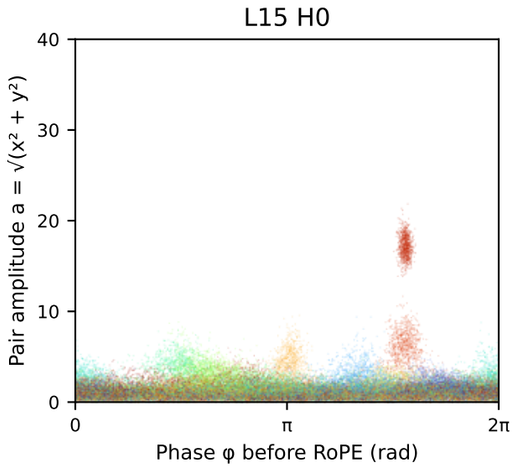
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

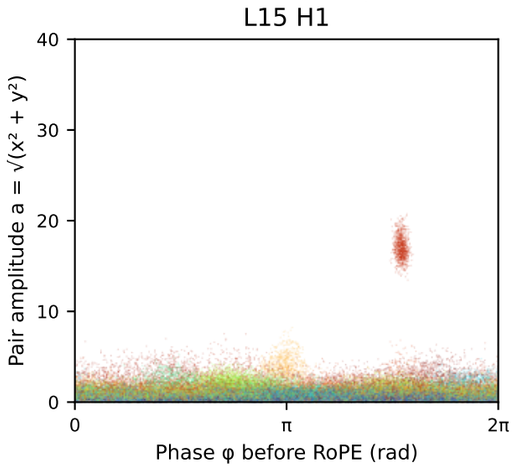
Qwen queries · layer 14 · amplitude axis 0–40

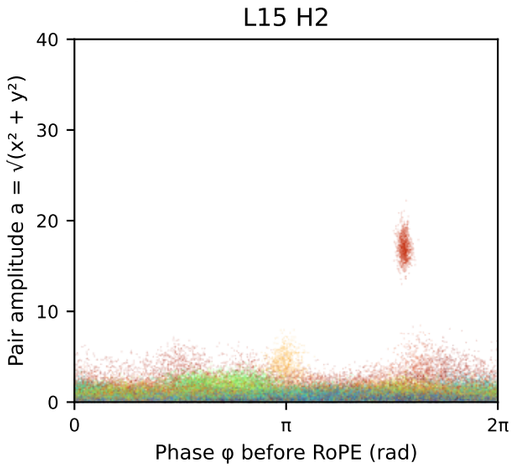


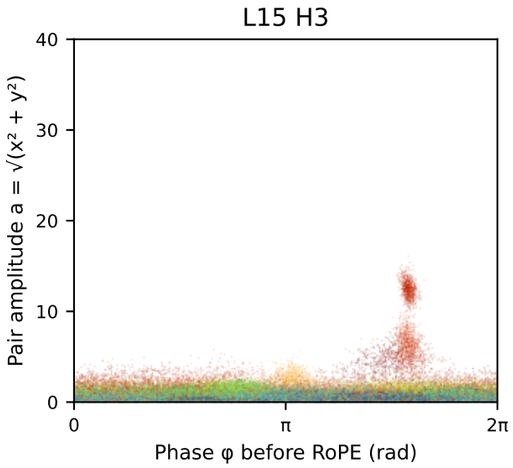
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

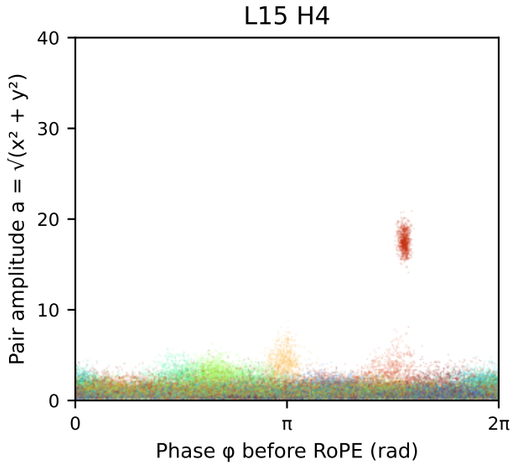
Qwen queries · layer 15 · amplitude axis 0–40

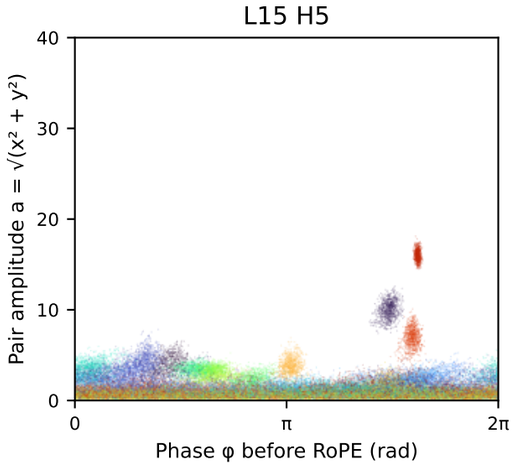


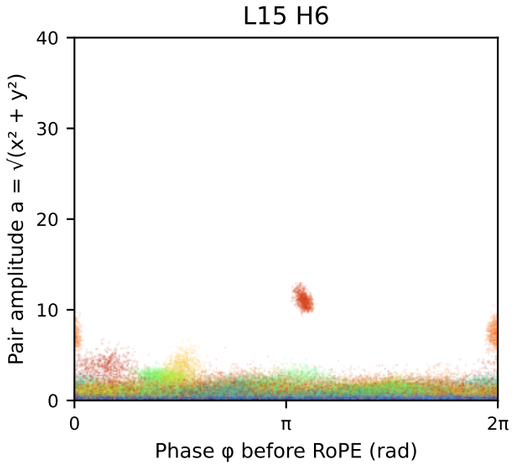


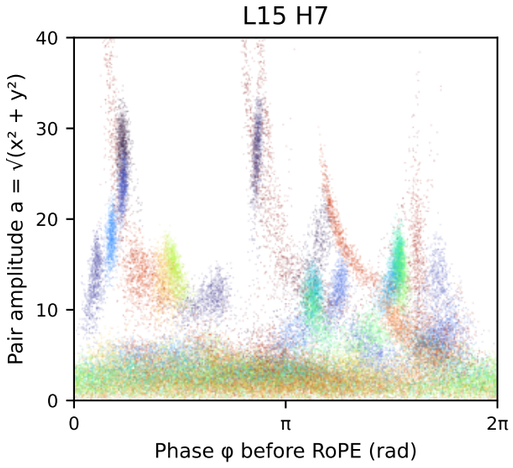


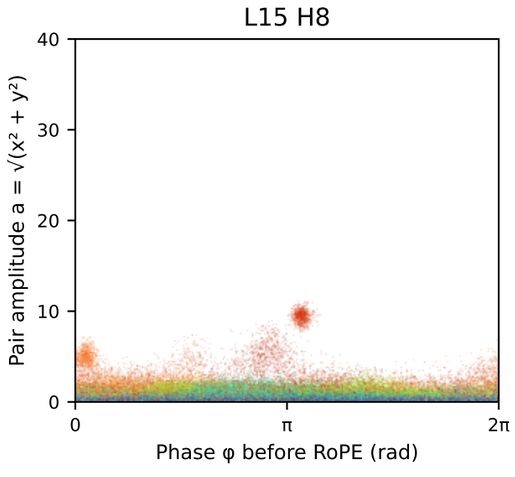


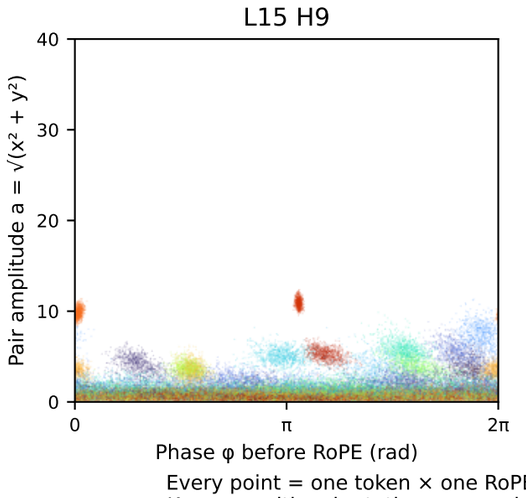


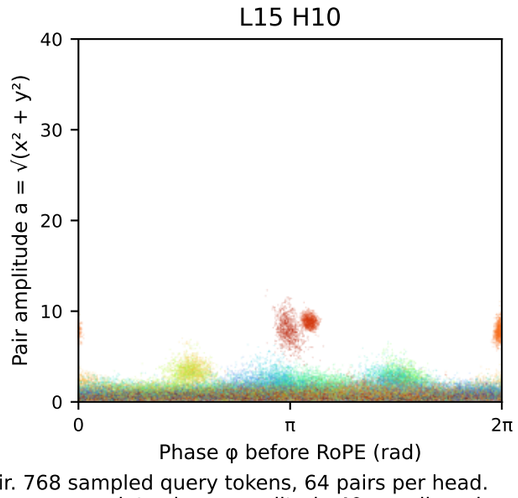


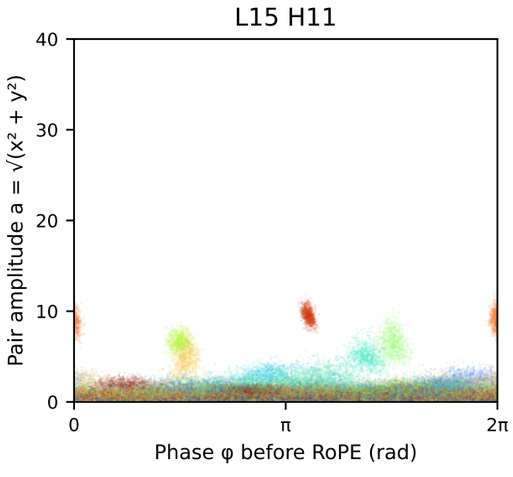


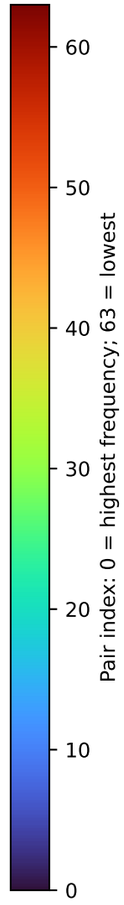








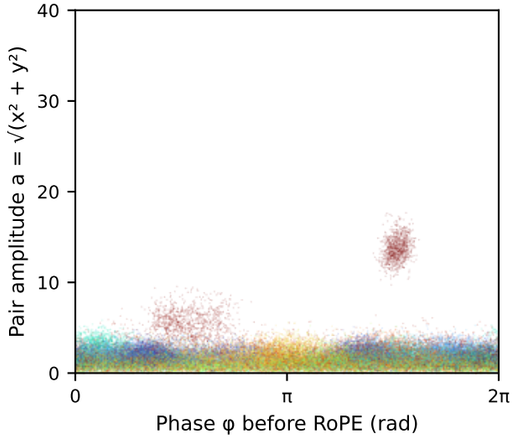




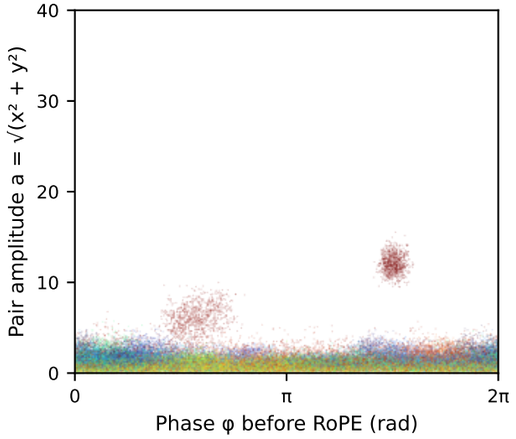
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 16 · amplitude axis 0–40

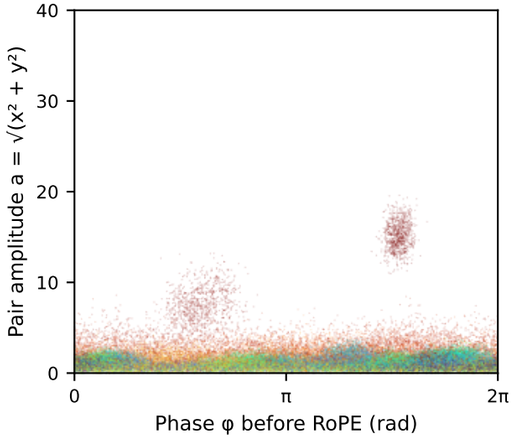
L16 H0



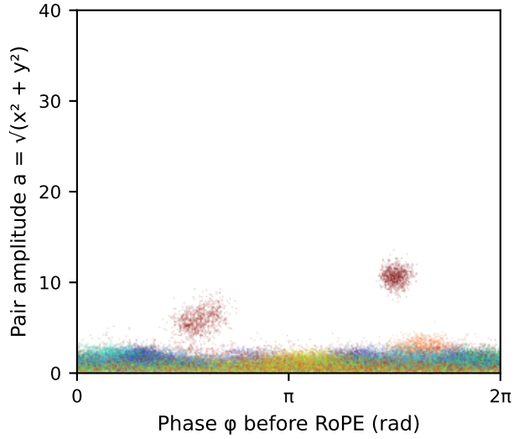
L16 H1



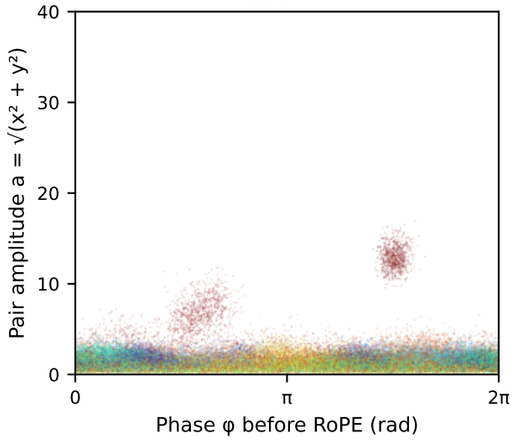
L16 H2



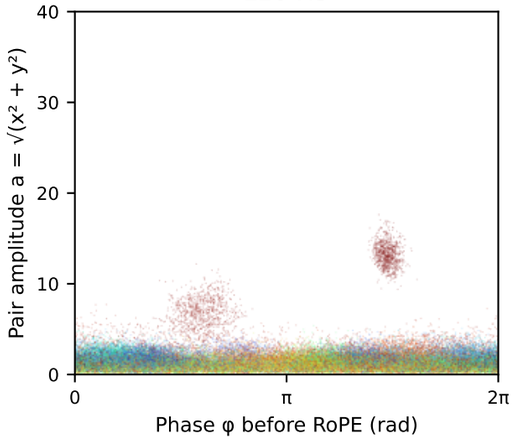
L16 H3



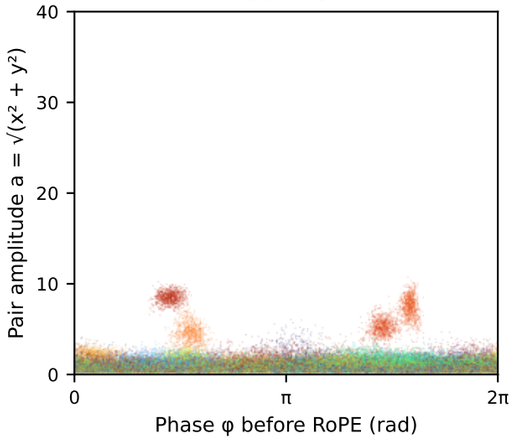
L16 H4



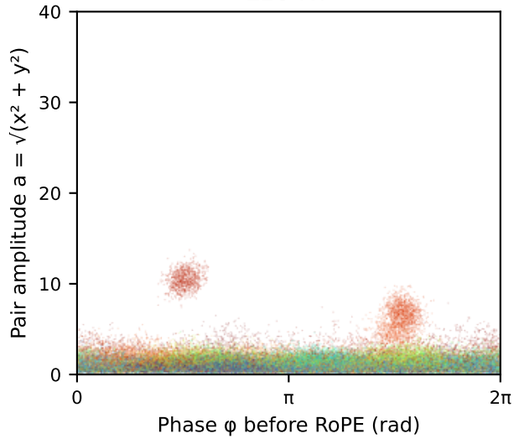
L16 H5



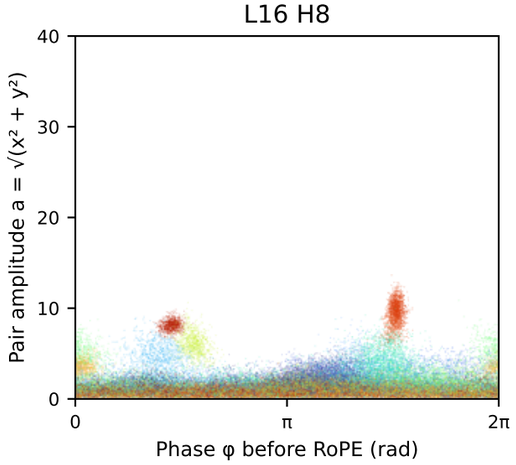
L16 H6



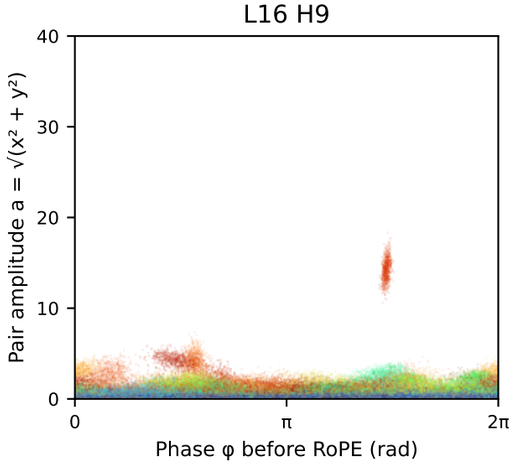
L16 H7



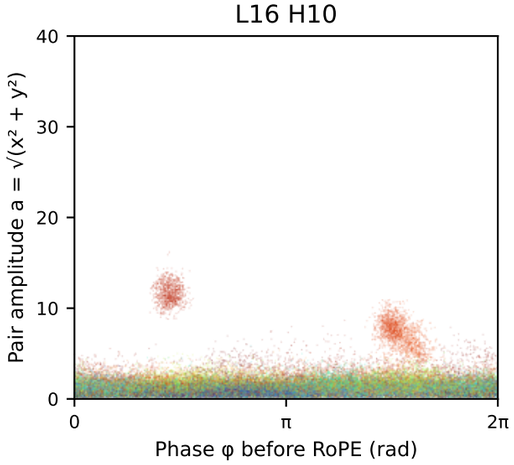
L16 H8



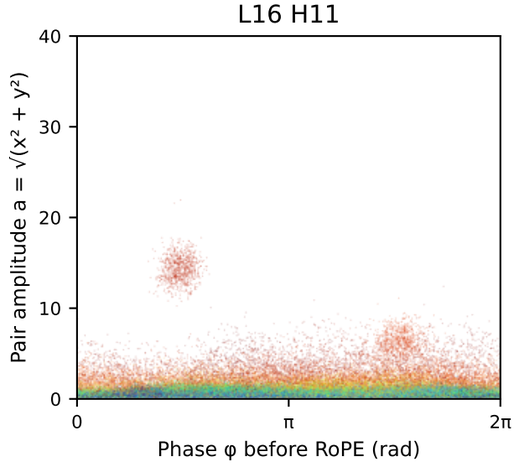
L16 H9

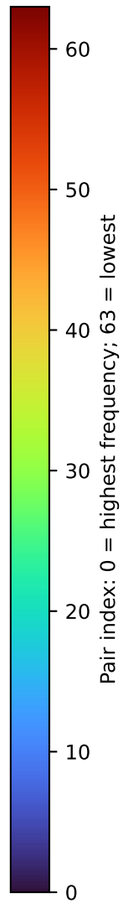


L16 H10



L16 H11





Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 17 · amplitude axis 0–40

L17 H0 | L17 H1 | L17 H2 | L17 H3

L17 H4 | L17 H5 | L17 H6 | L17 H7

L17 H8 | L17 H9 | L17 H10 | L17 H11

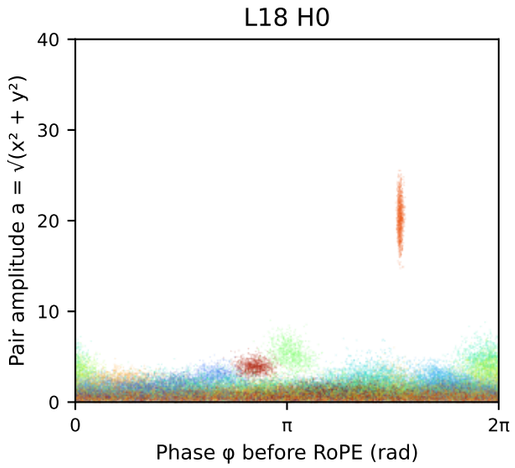
Pair amplitude a = √(x² + y²)

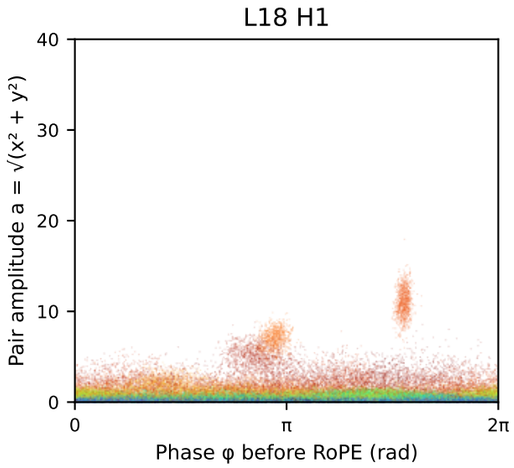
Phase φ before RoPE (rad)

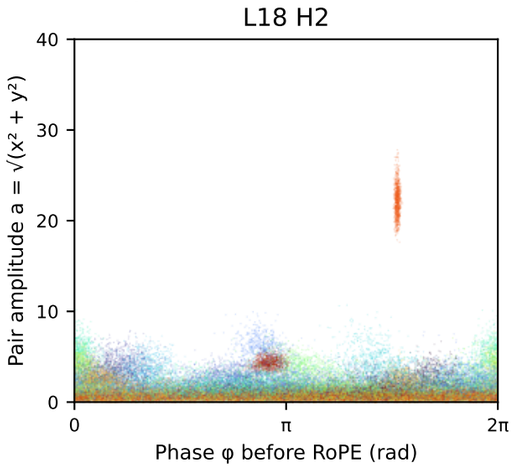
Pair index: 0 = highest frequency; 63 = lowest

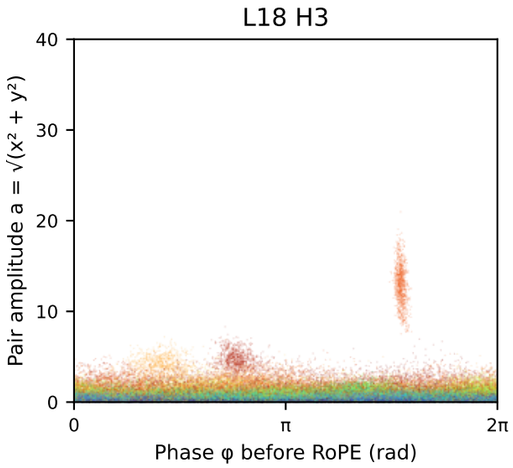
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

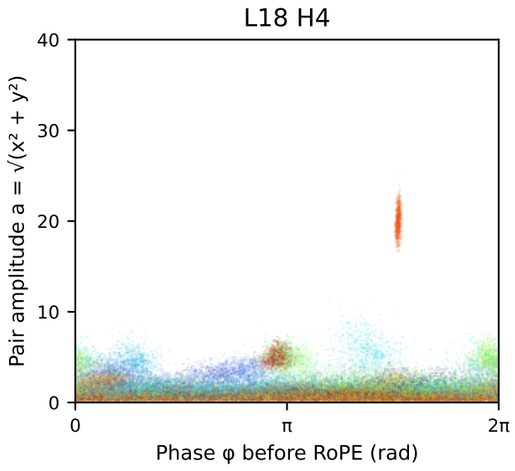
Qwen queries · layer 18 · amplitude axis 0–40

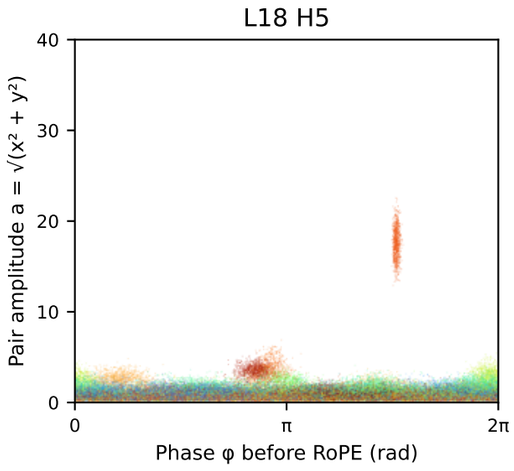


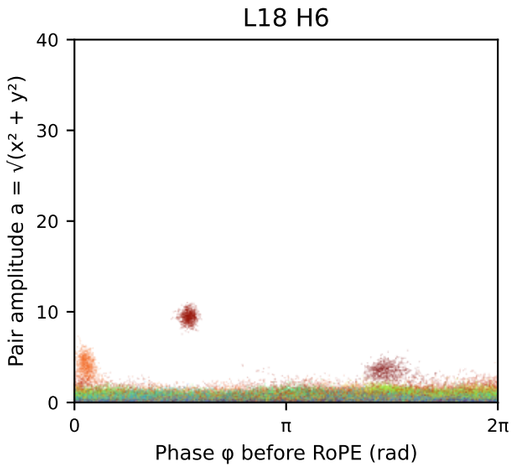


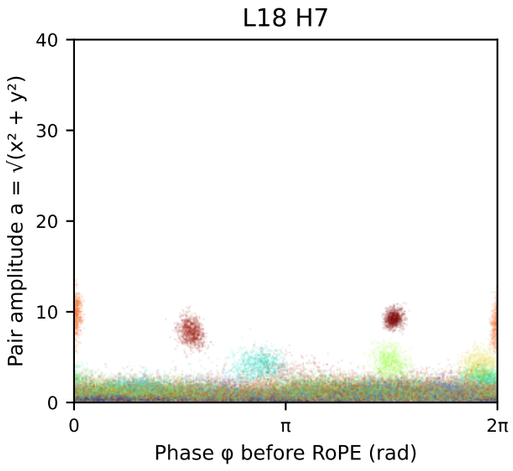


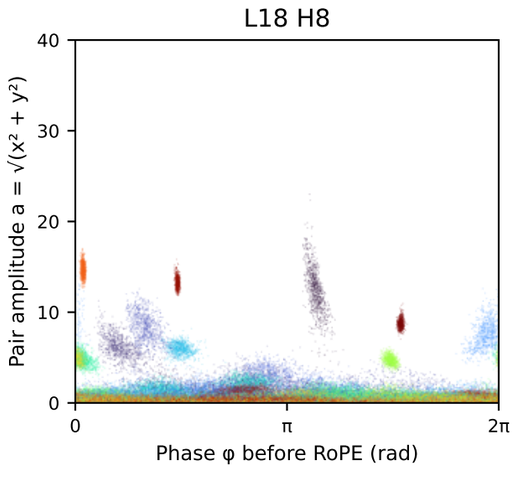


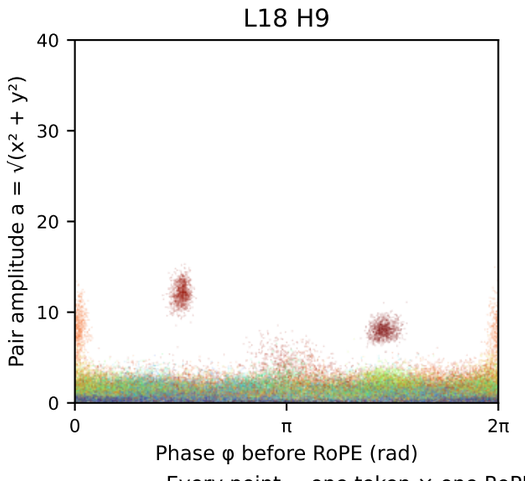


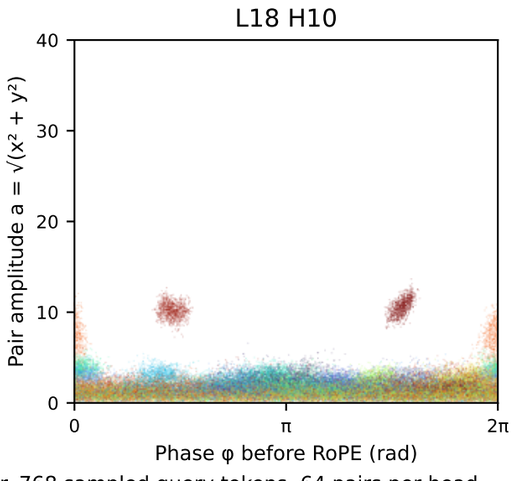


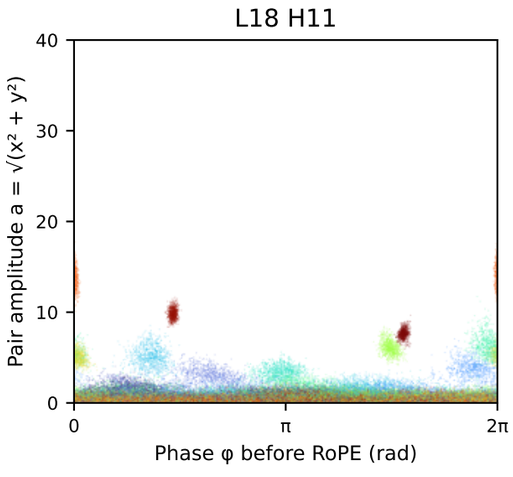


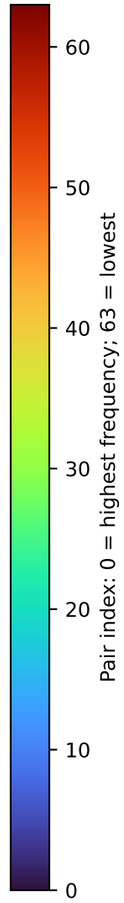






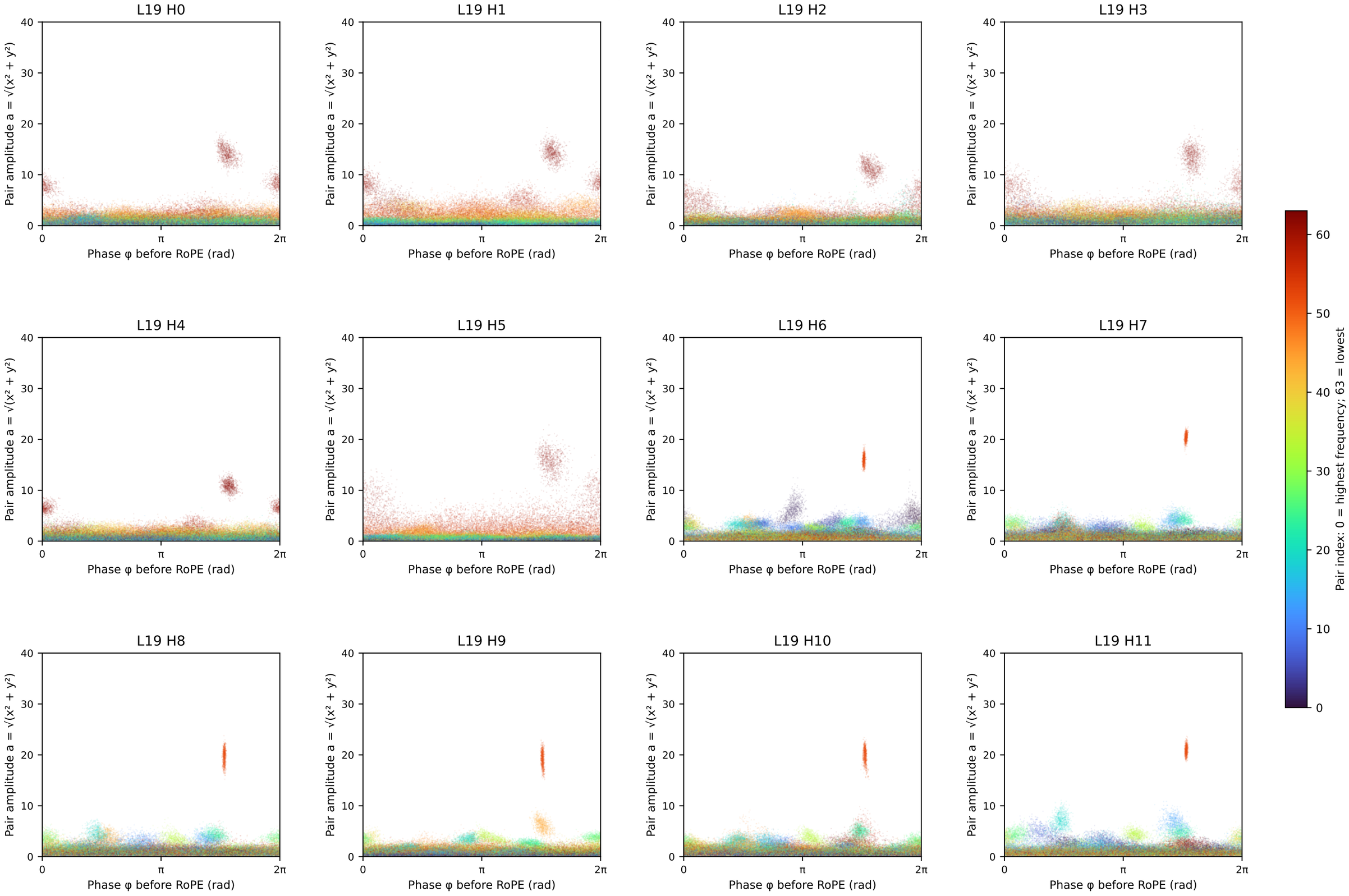






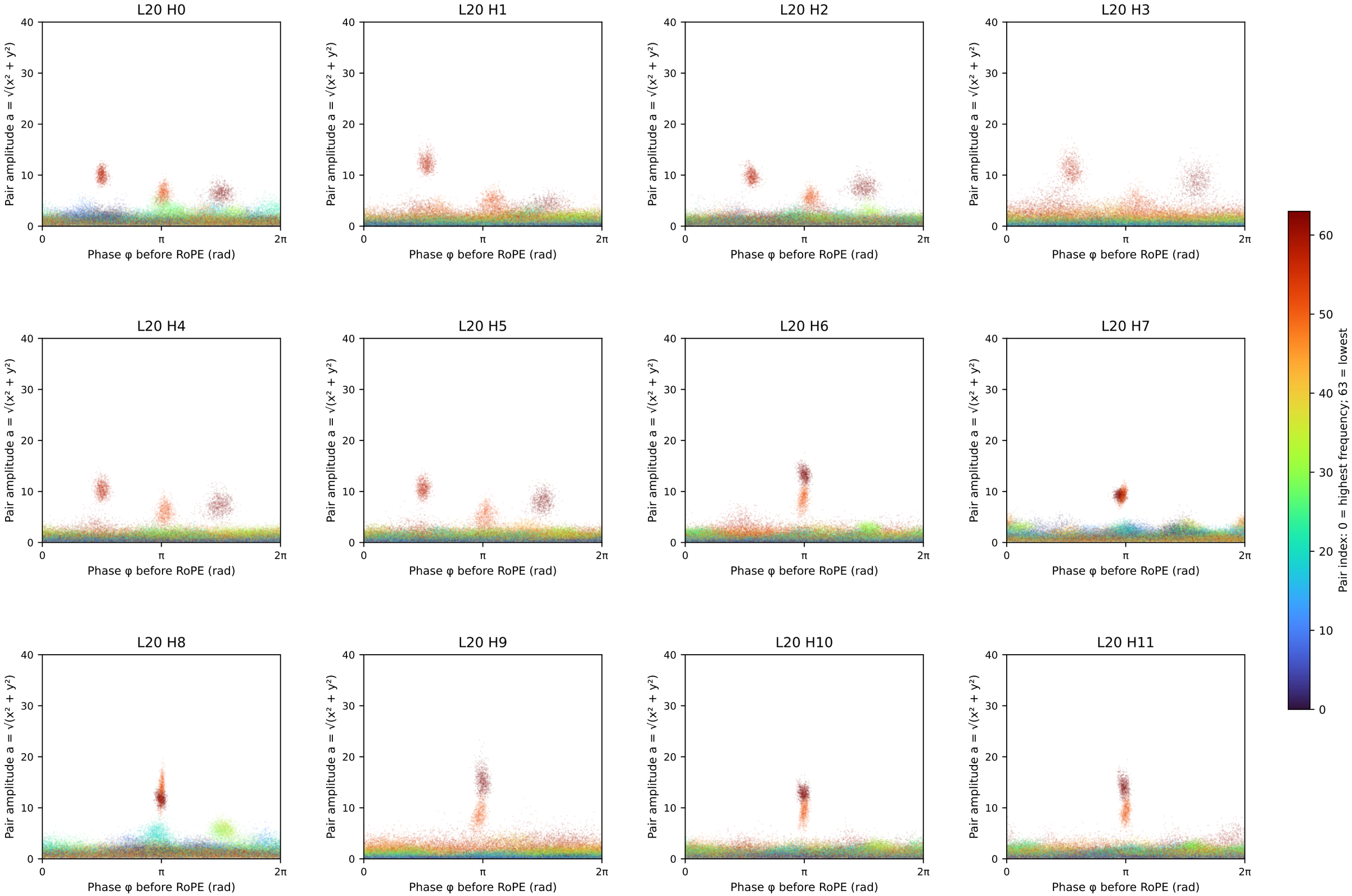
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 19 · amplitude axis 0–40



Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

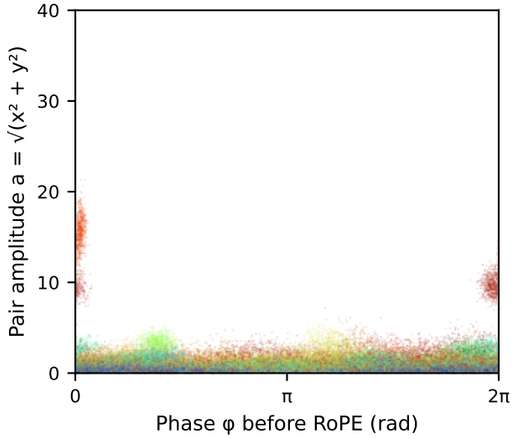
Qwen queries · layer 20 · amplitude axis 0–40



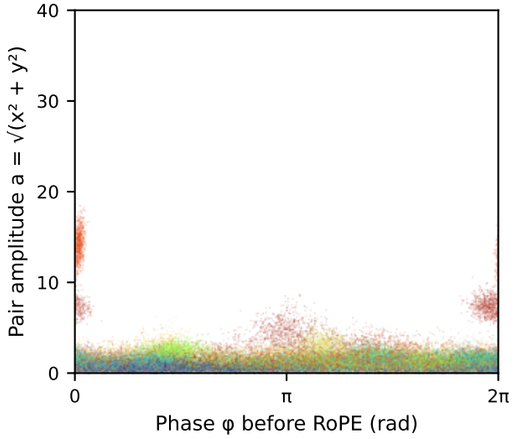
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 21 · amplitude axis 0–40

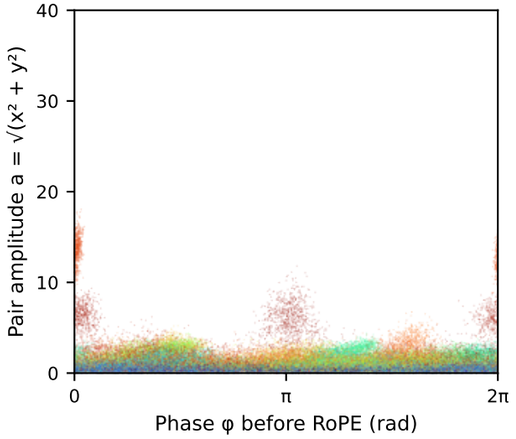
L21 H0



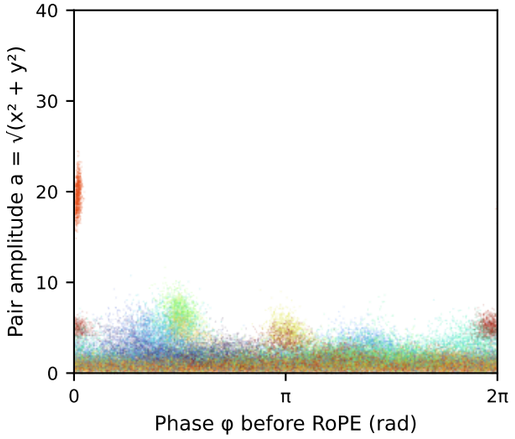
L21 H1



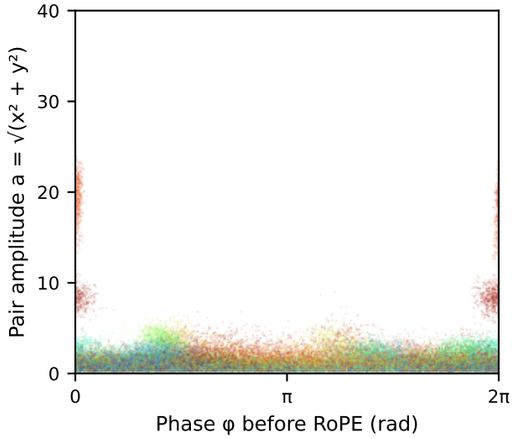
L21 H2



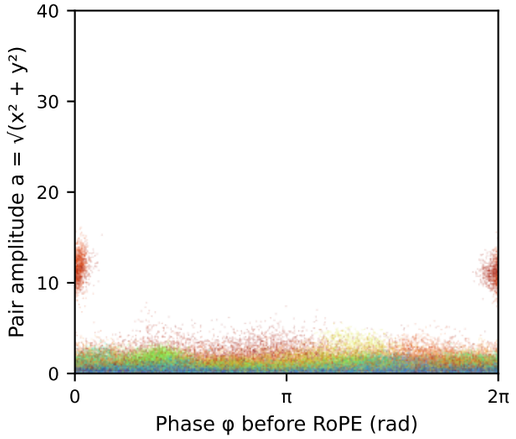
L21 H3



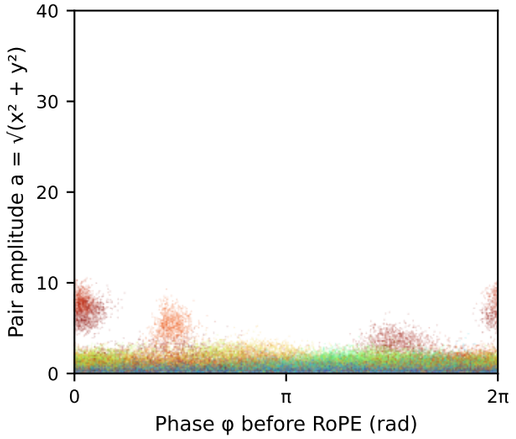
L21 H4



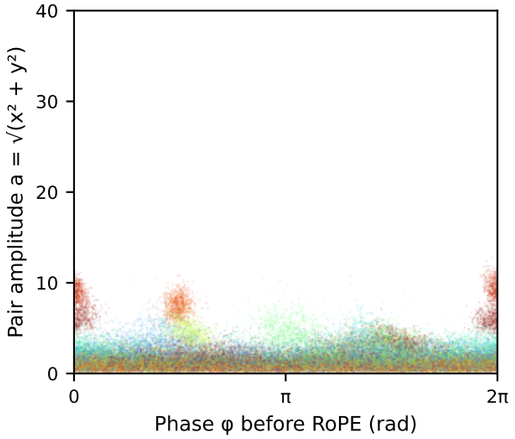
L21 H5



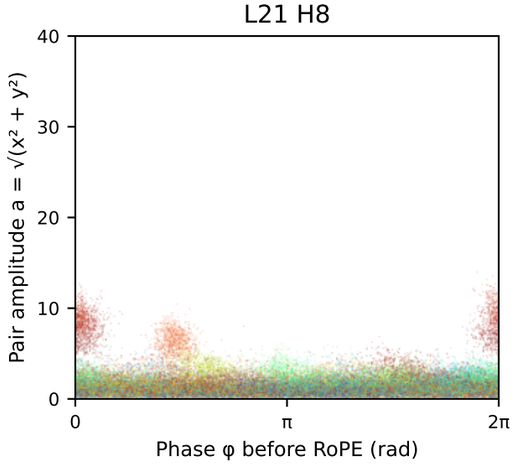
L21 H6



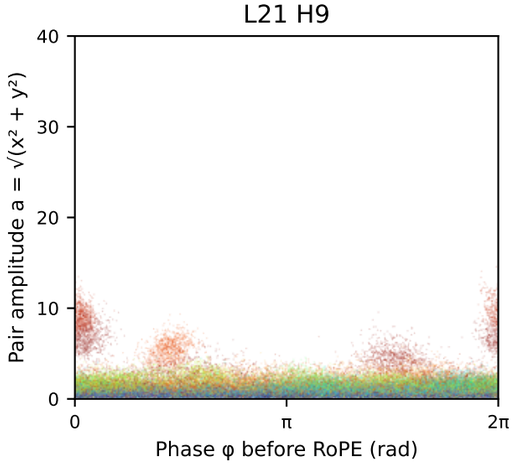
L21 H7



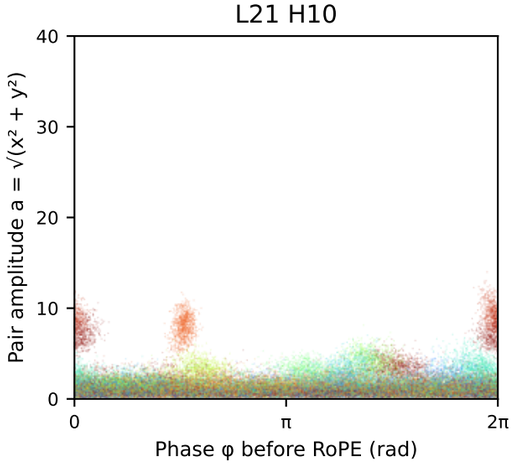
L21 H8



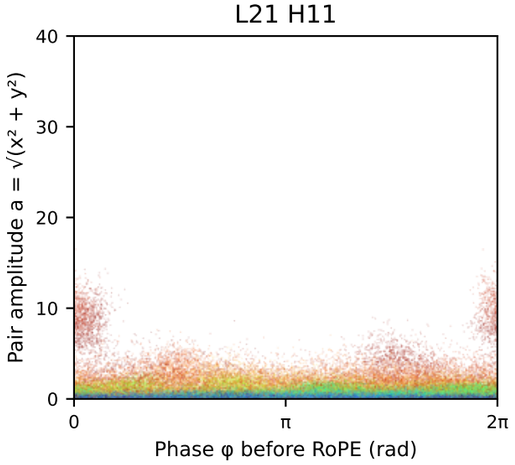
L21 H9

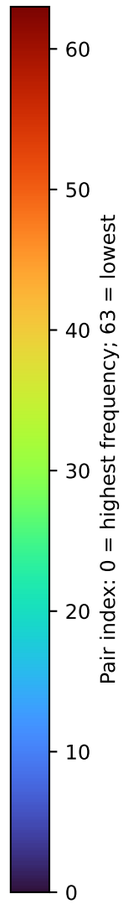


L21 H10



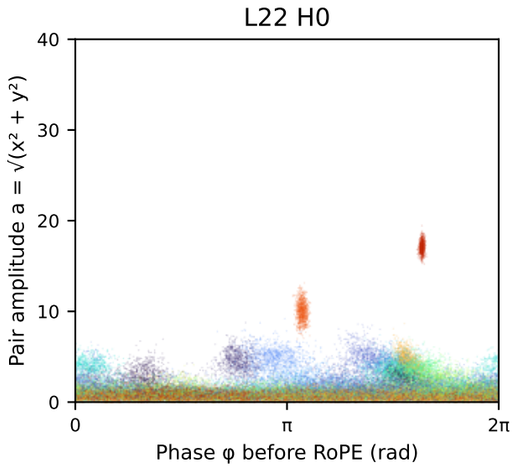
L21 H11

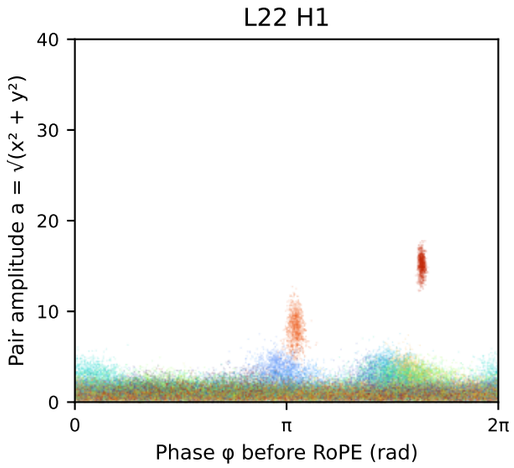


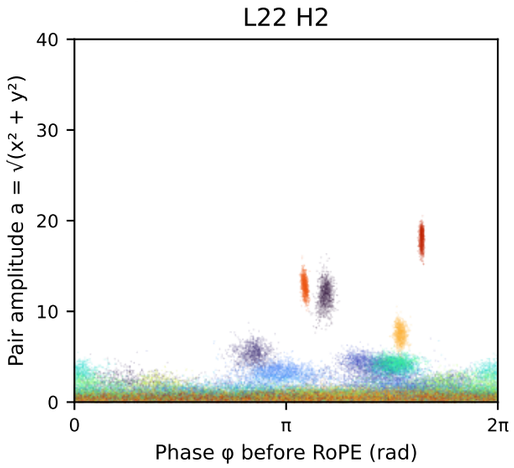


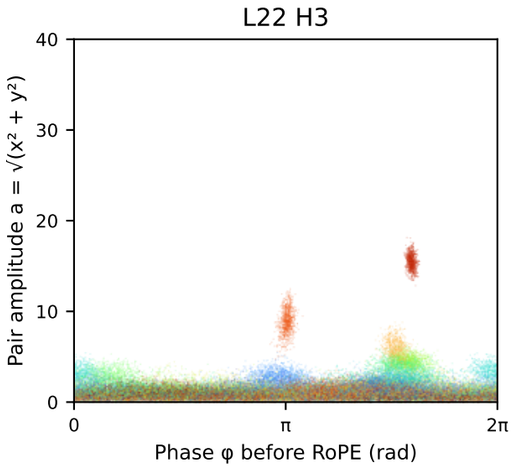
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

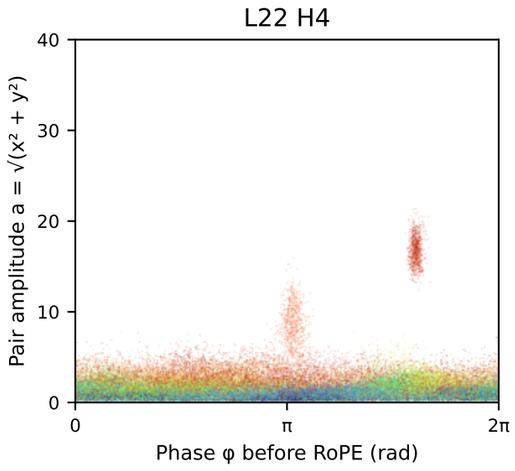
Qwen queries · layer 22 · amplitude axis 0–40

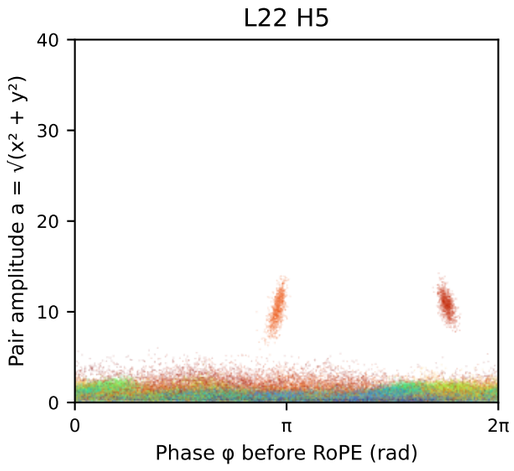


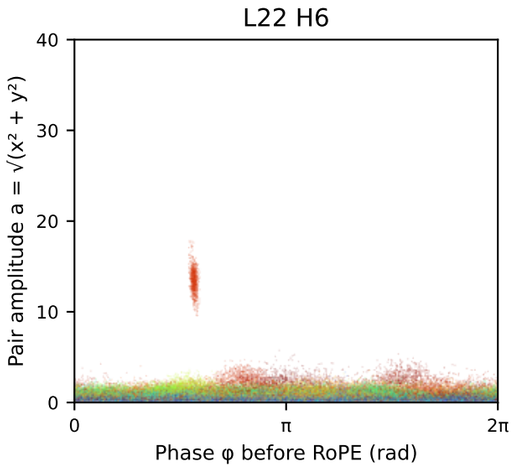


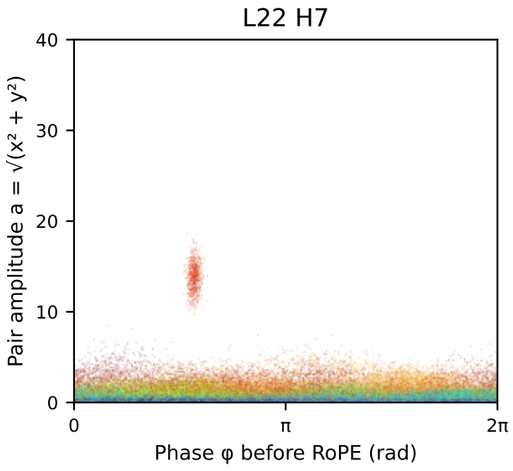


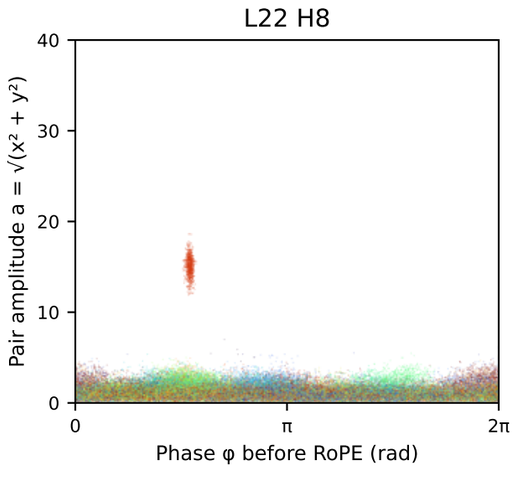


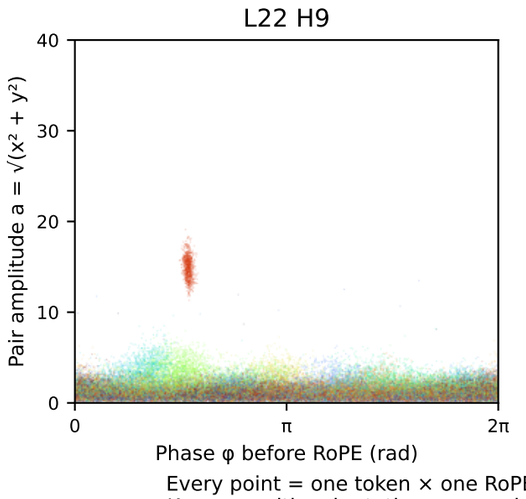


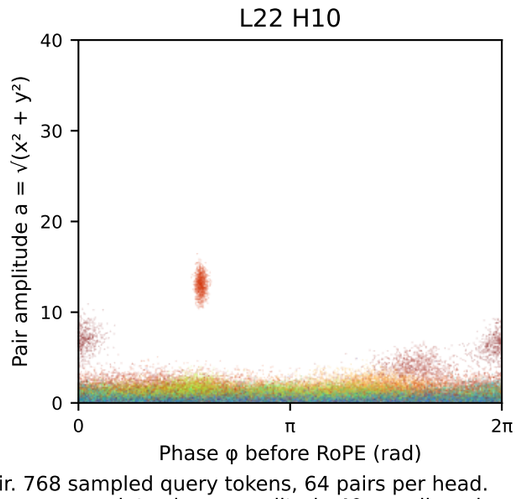


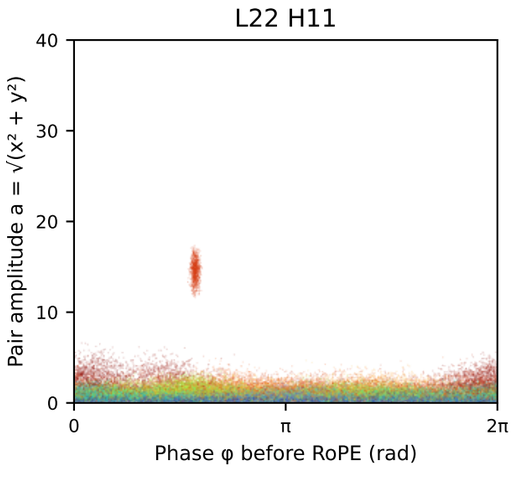


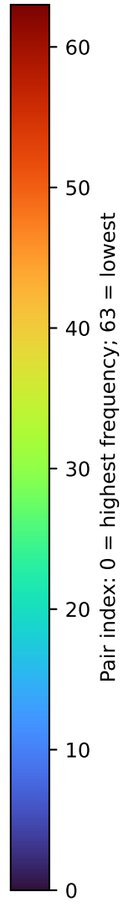






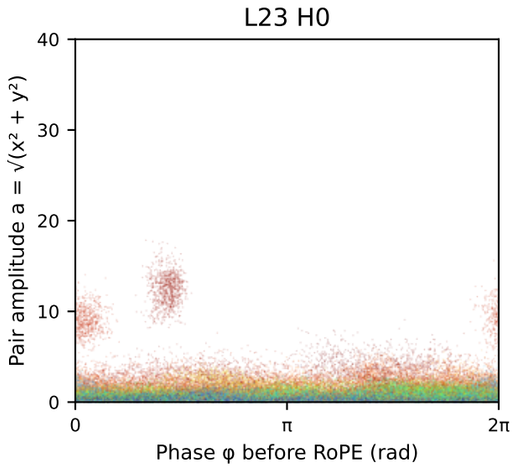


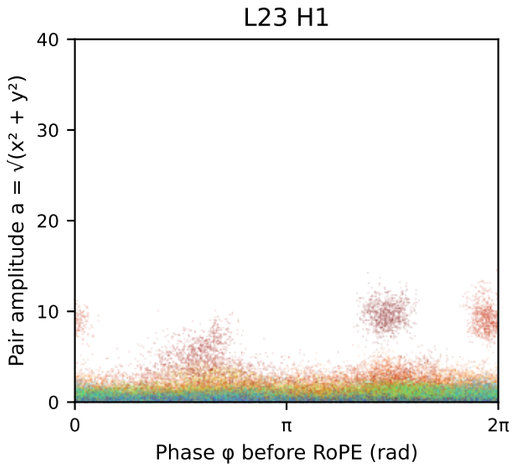


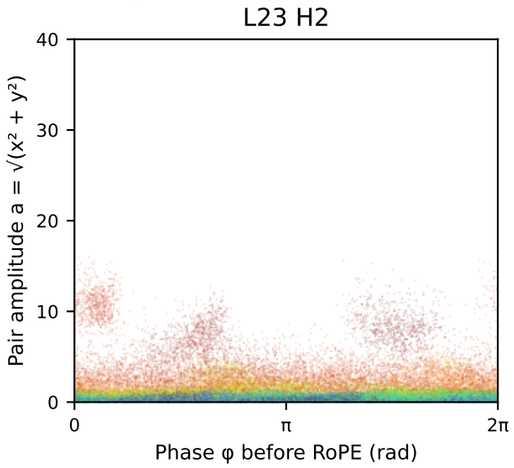


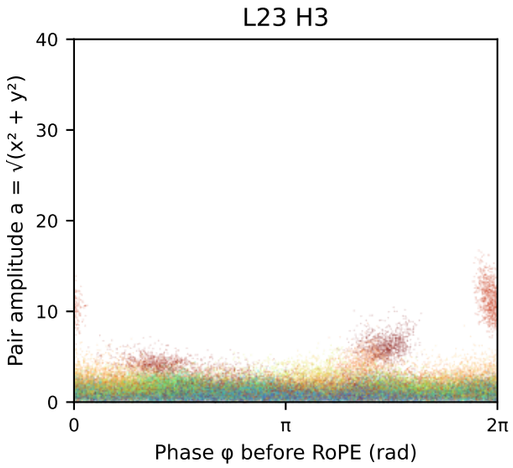
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

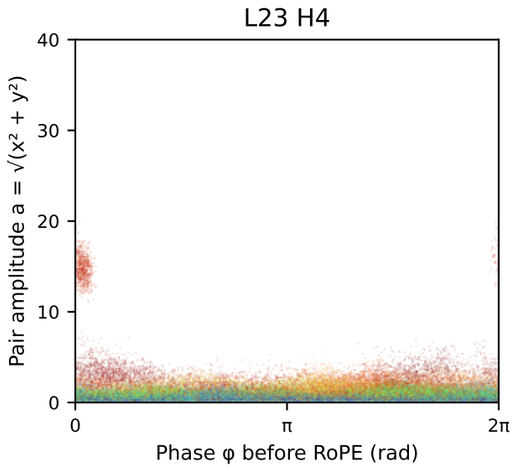
Qwen queries · layer 23 · amplitude axis 0–40

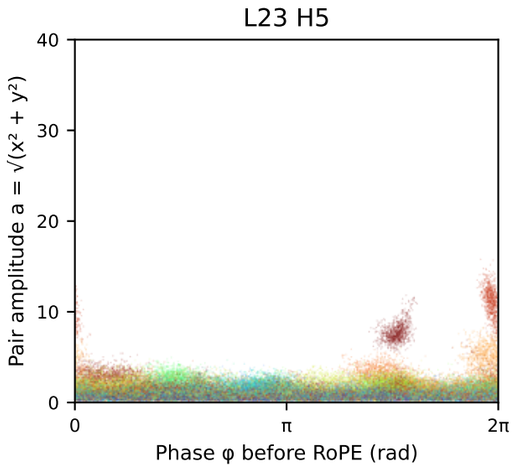


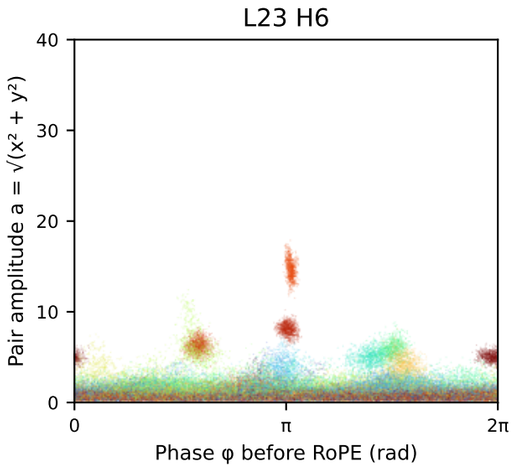


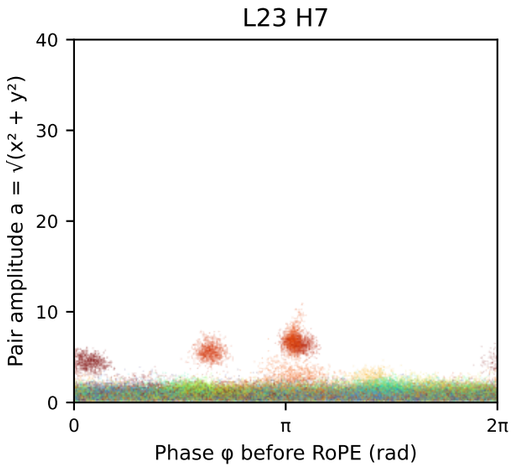


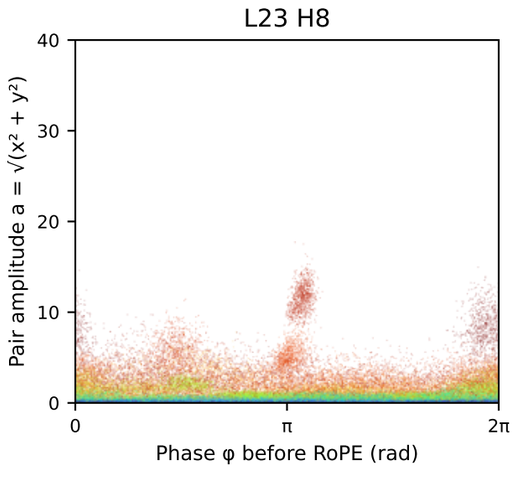


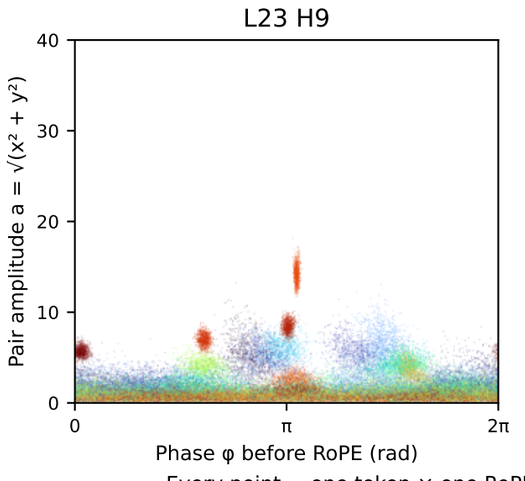


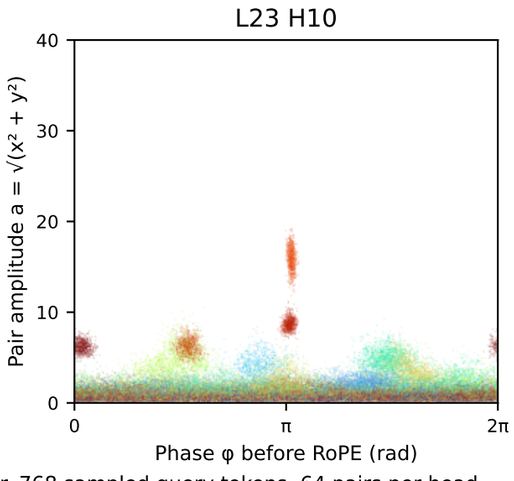


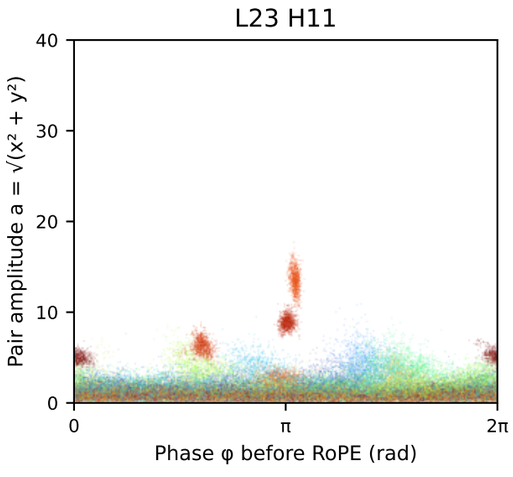


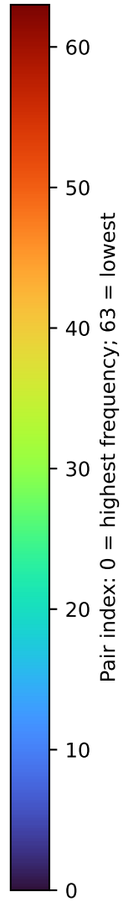






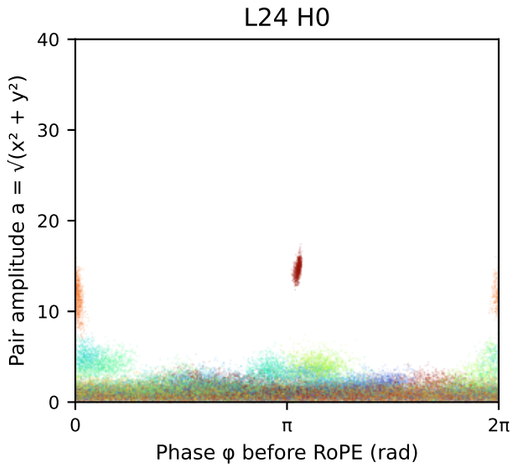


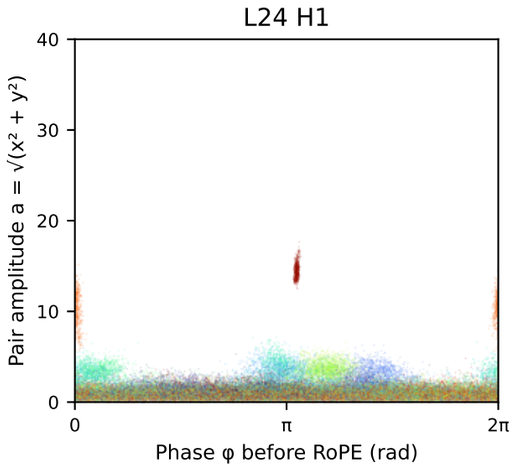


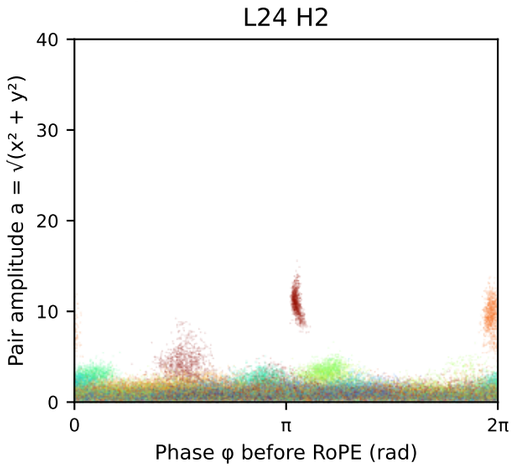


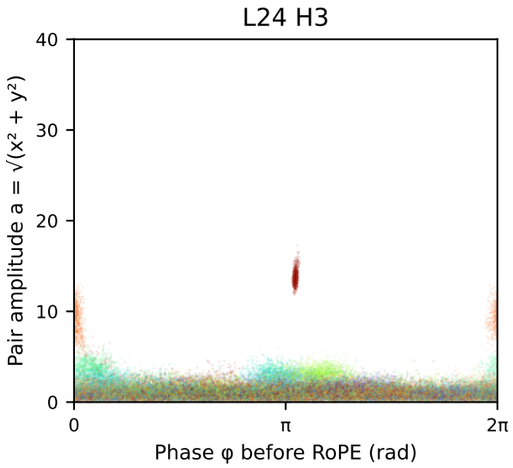
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

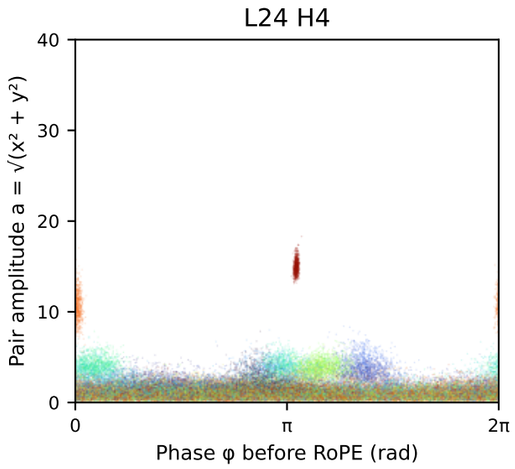
Qwen queries · layer 24 · amplitude axis 0–40

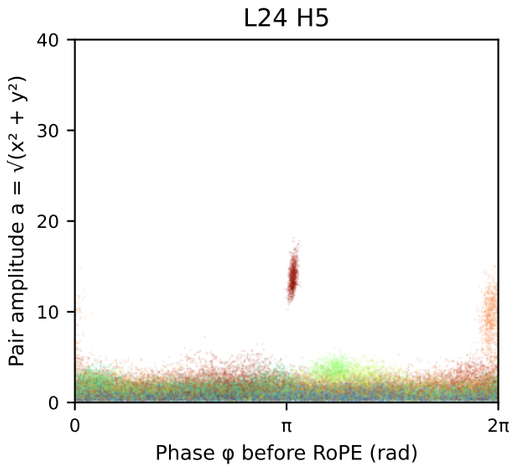


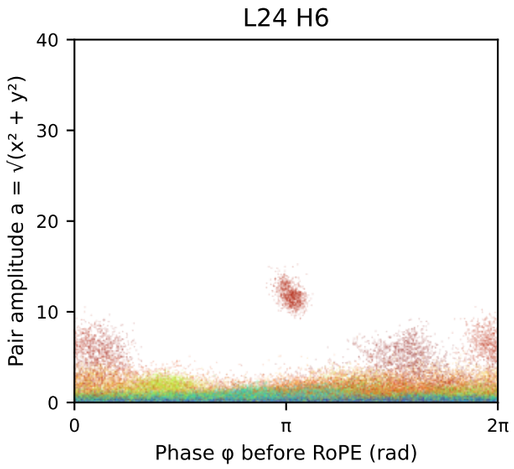


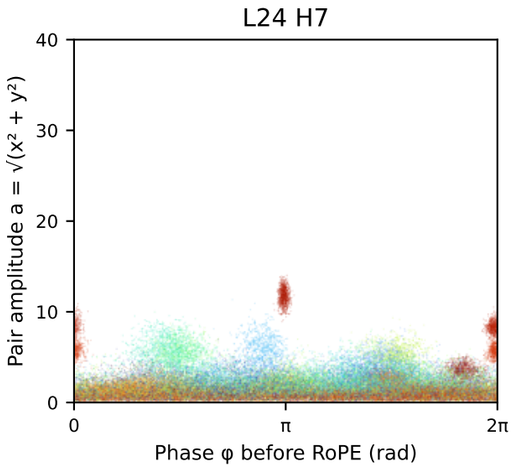


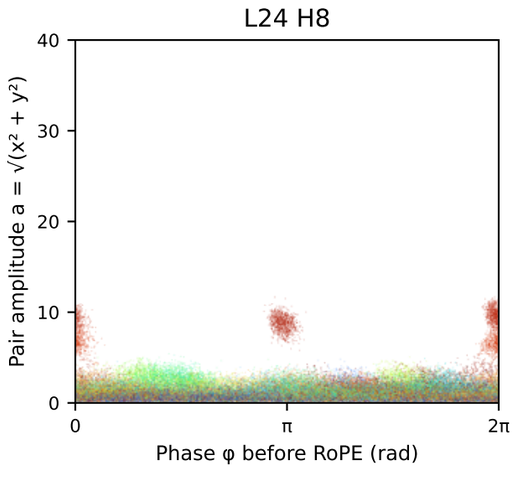


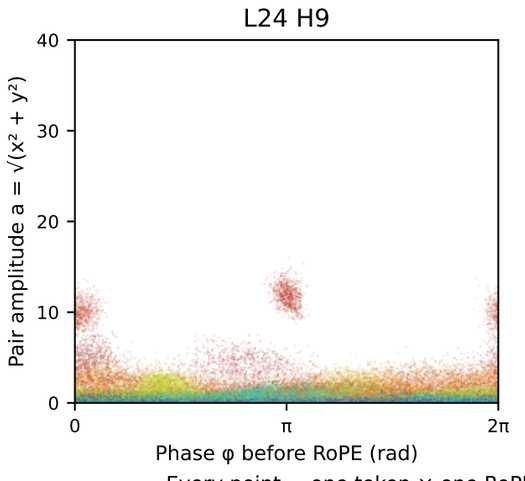


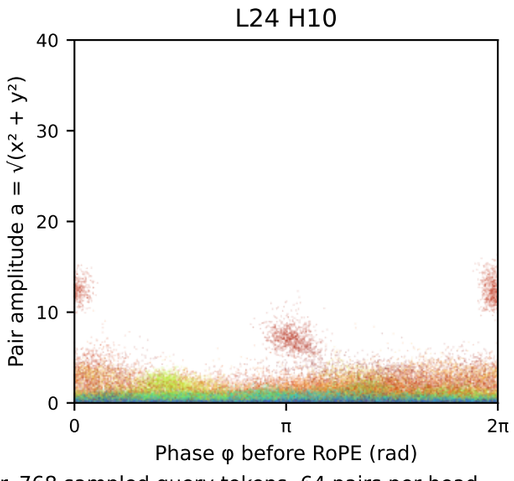


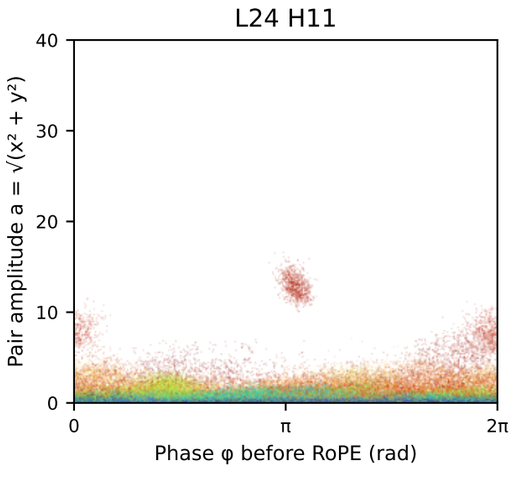


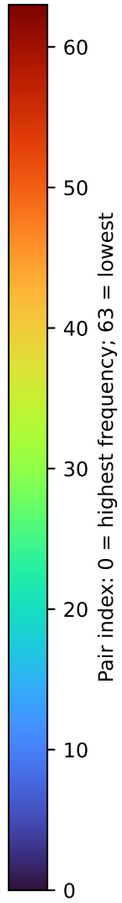












Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 25 · amplitude axis 0–40

L25 H0 | L25 H1 | L25 H2 | L25 H3

L25 H4 | L25 H5 | L25 H6 | L25 H7

L25 H8 | L25 H9 | L25 H10 | L25 H11

Pair amplitude a = √(x² + y²)

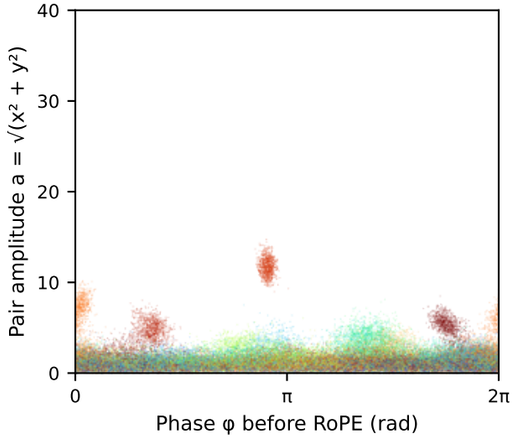
Phase φ before RoPE (rad)

Pair index: 0 = highest frequency; 63 = lowest

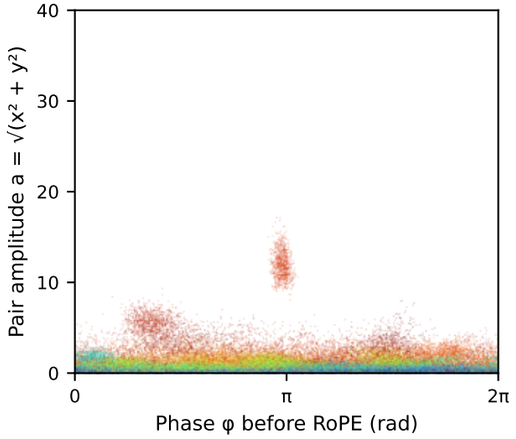
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 26 · amplitude axis 0–40

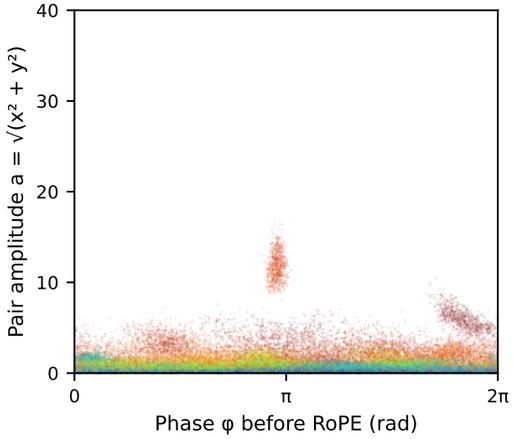
L26 H0



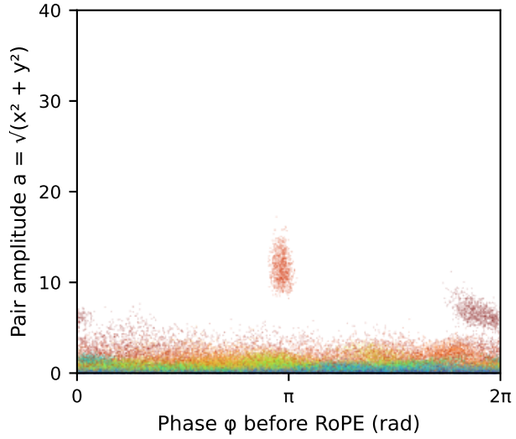
L26 H1



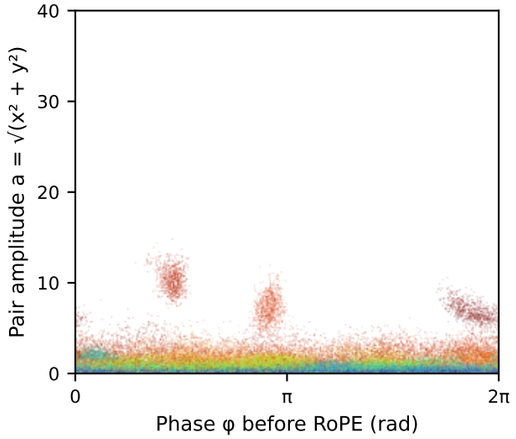
L26 H2



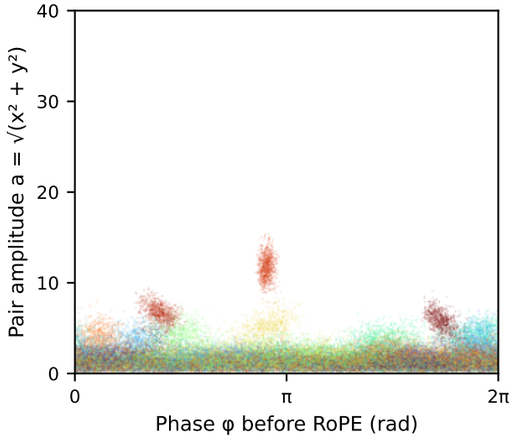
L26 H3



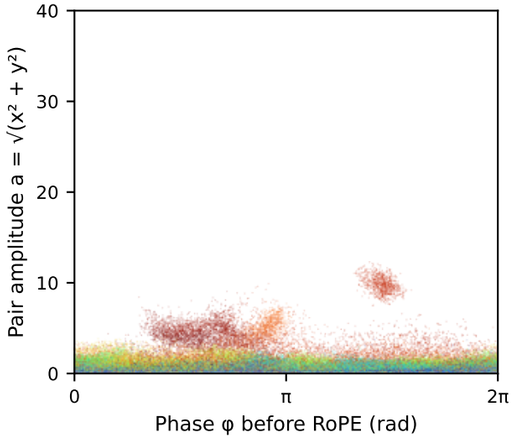
L26 H4



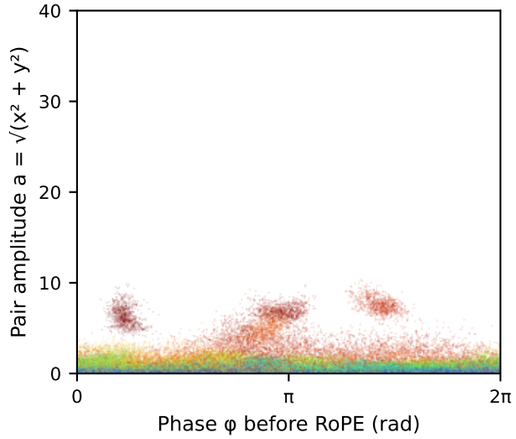
L26 H5



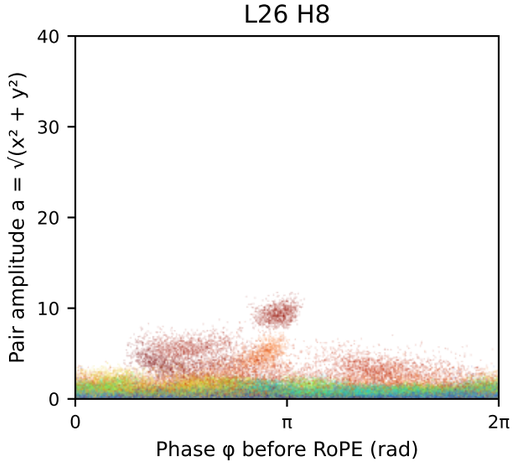
L26 H6



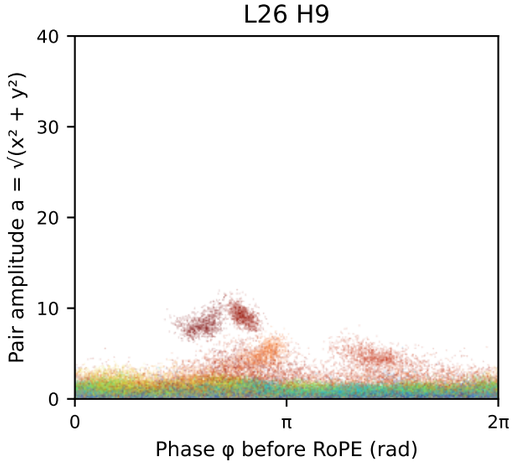
L26 H7



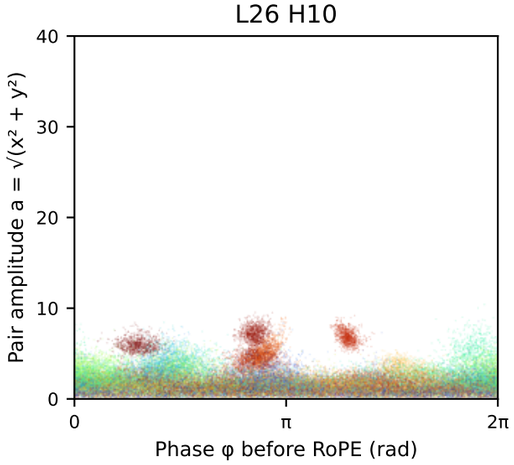
L26 H8



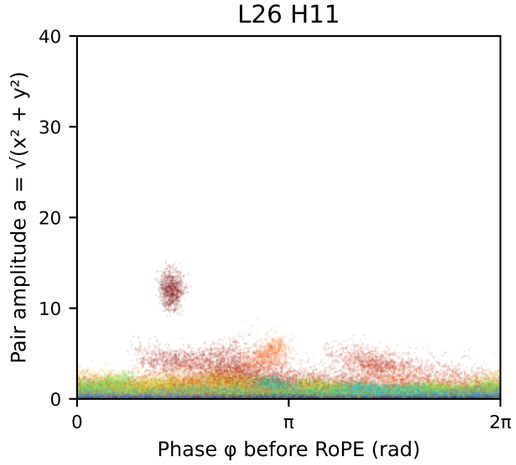
L26 H9

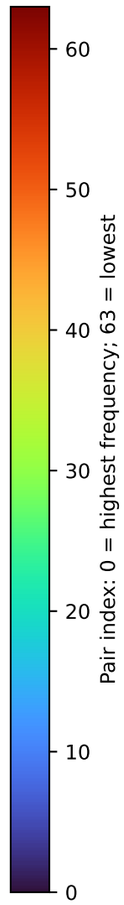


L26 H10



L26 H11

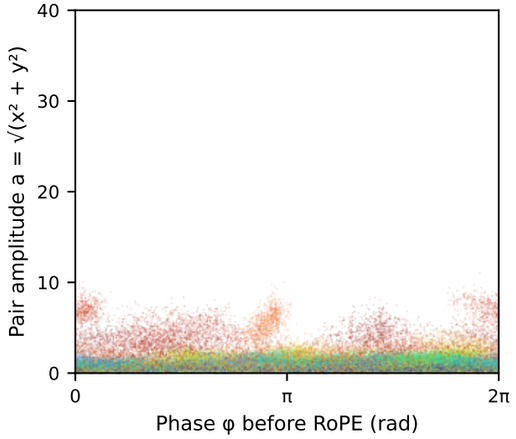




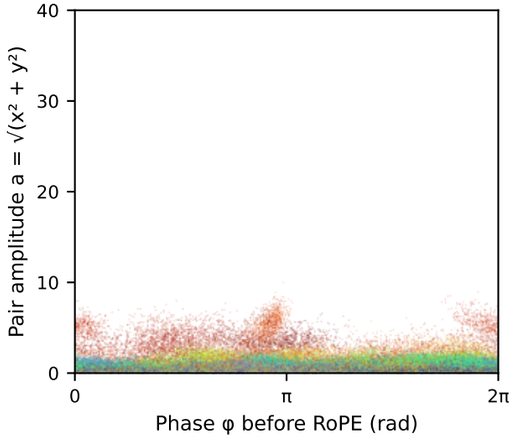
Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.

Qwen queries · layer 27 · amplitude axis 0–40

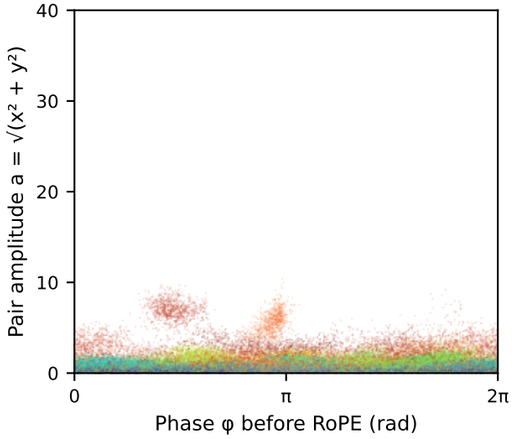
L27 H0



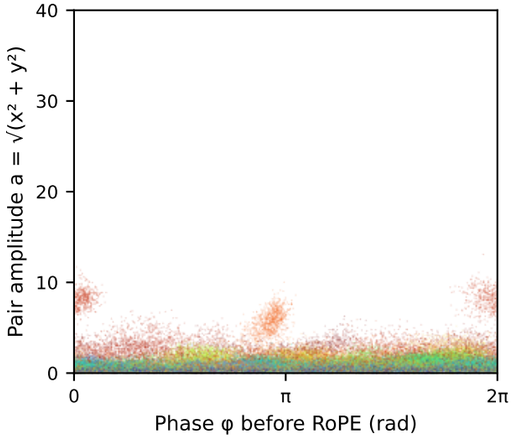
L27 H1



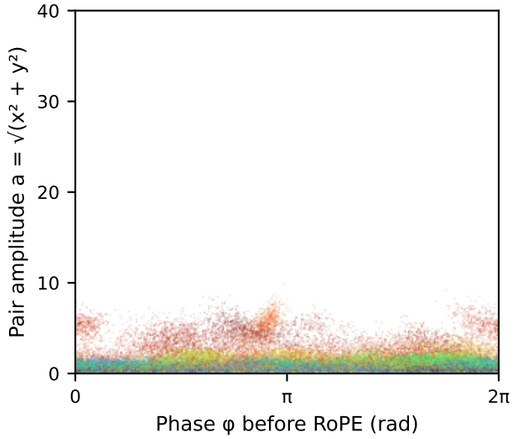
L27 H2



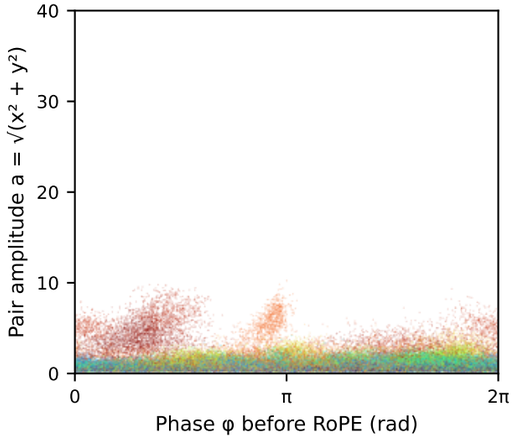
L27 H3



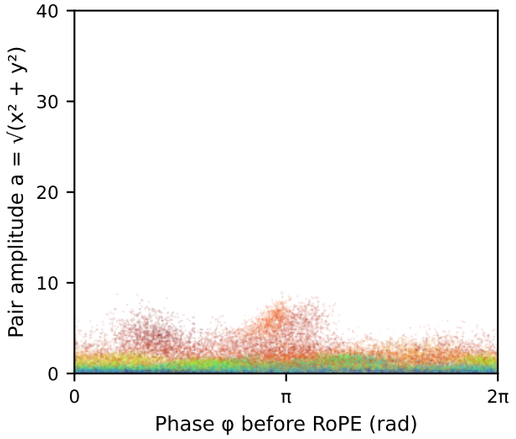
L27 H4



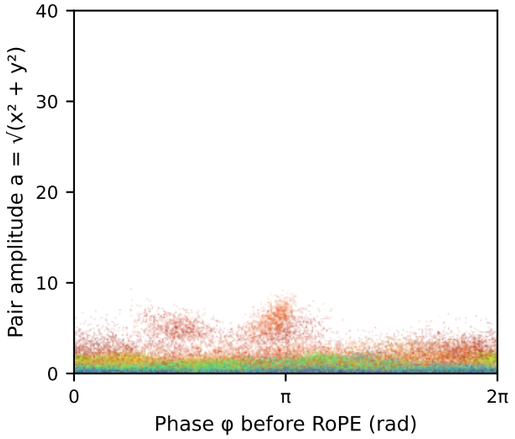
L27 H5



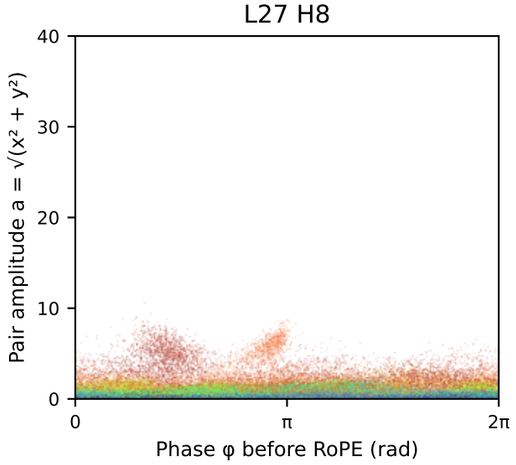
L27 H6



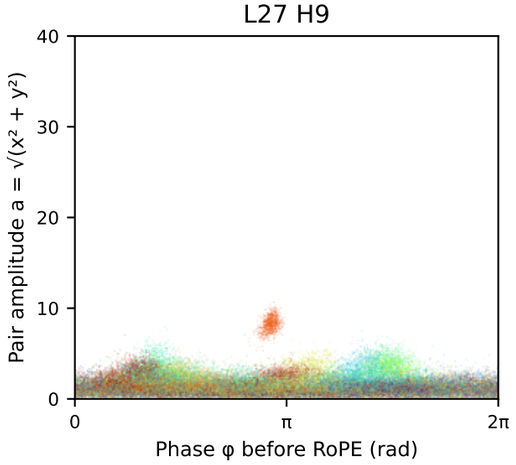
L27 H7



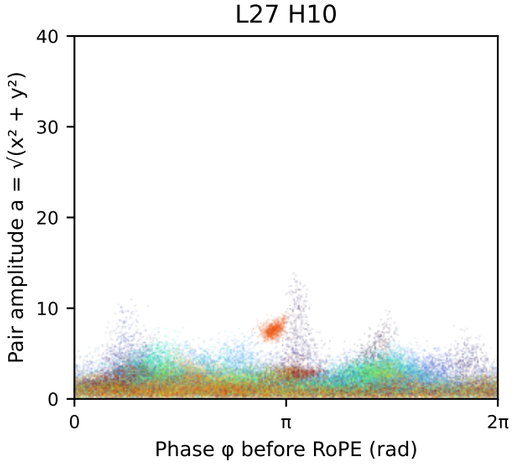
L27 H8



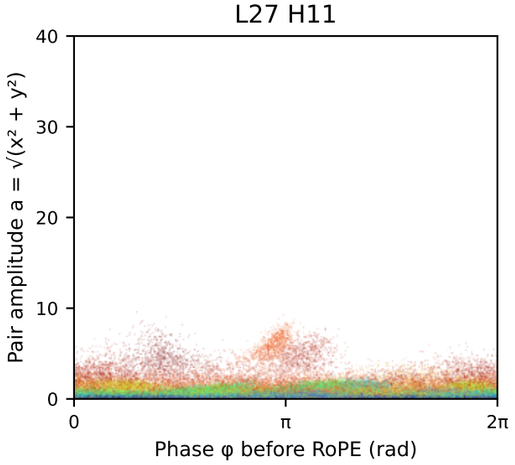
L27 H9

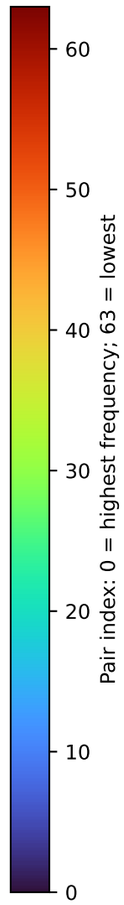


L27 H10



L27 H11





Every point = one token × one RoPE pair. 768 sampled query tokens, 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Native query vectors only; no key or margin enters this plot.